Departamento de
CIÊNCIA DA COMPUTAÇÃO

Prof. Filipe Dwan Pereira

INTELIGÊNCIA ARTIFICIAL

Otimização por colônia de formigas

Disclaimer:

Esta aula é baseada nos slides e materiais do professor Guilherme Matos - IA Expert Academy

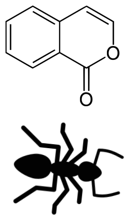
ANT COLONY OPTIMIZATION (ACO)

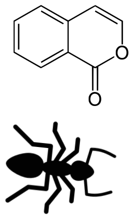
- Ant Colony Optimization (ACO)
- Algorithm proposed by Dorigo, in 1992
- It is based on the foraging behavior of ants in search for food
- Ants communicate with pheromones, and they tend to follow pheromone trails (stygmergy)
- Pheromone evaporates with time

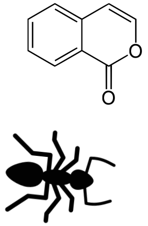
Applications

1.Internet routing
2.GPS systems
3.Edge detection in images

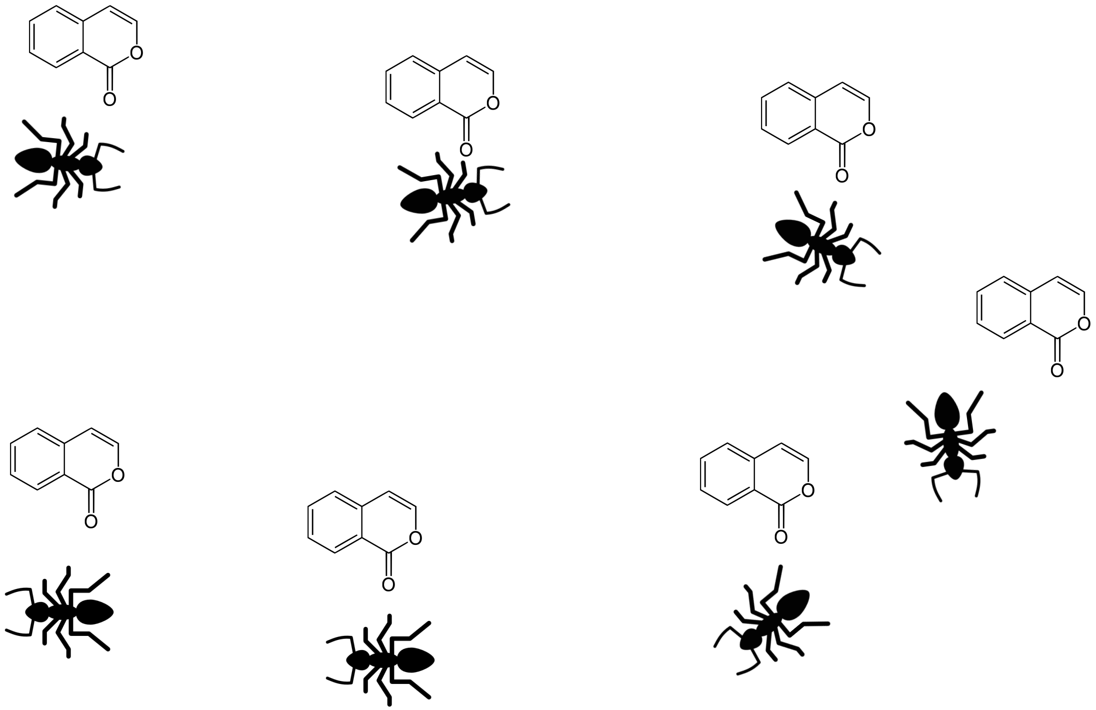
PHEROMONE RELEASE





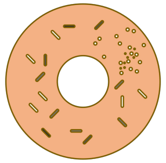


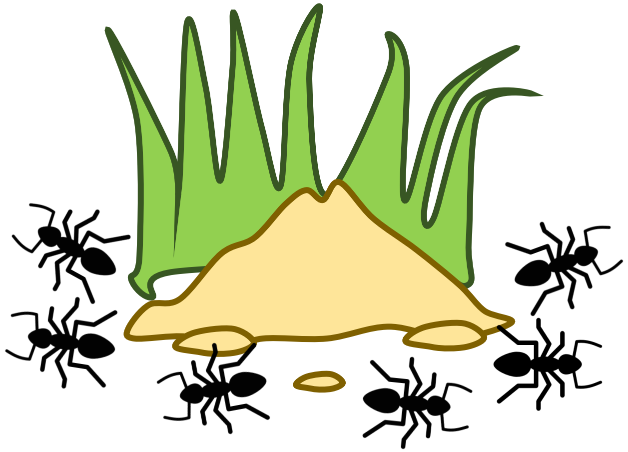
PHEROMONE EVAPORATION



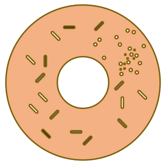
FOOD FORAGING BEHAVIOR

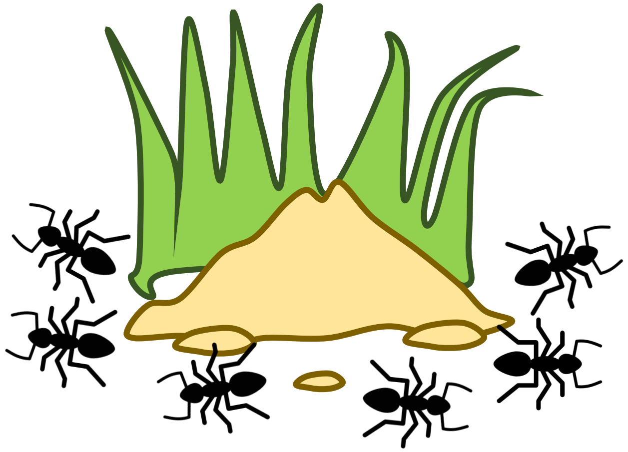
Food



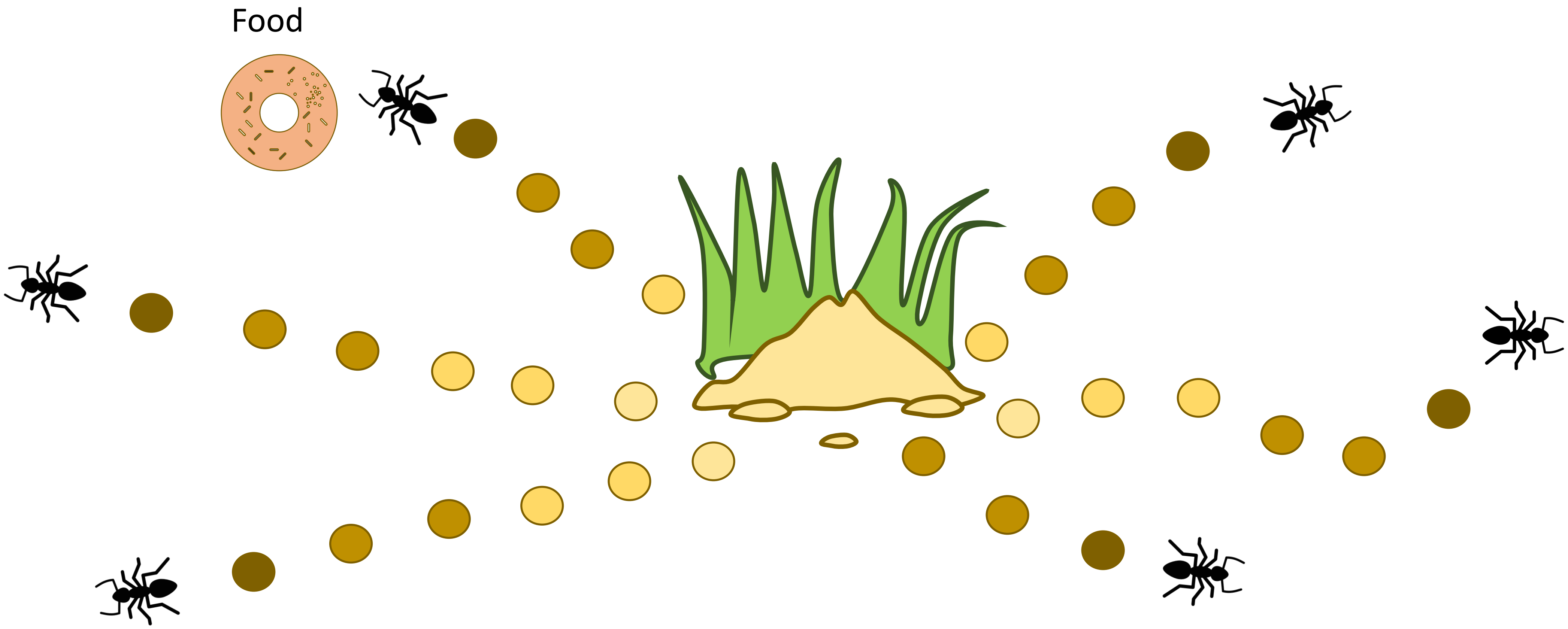


FOOD FORAGING BEHAVIOR

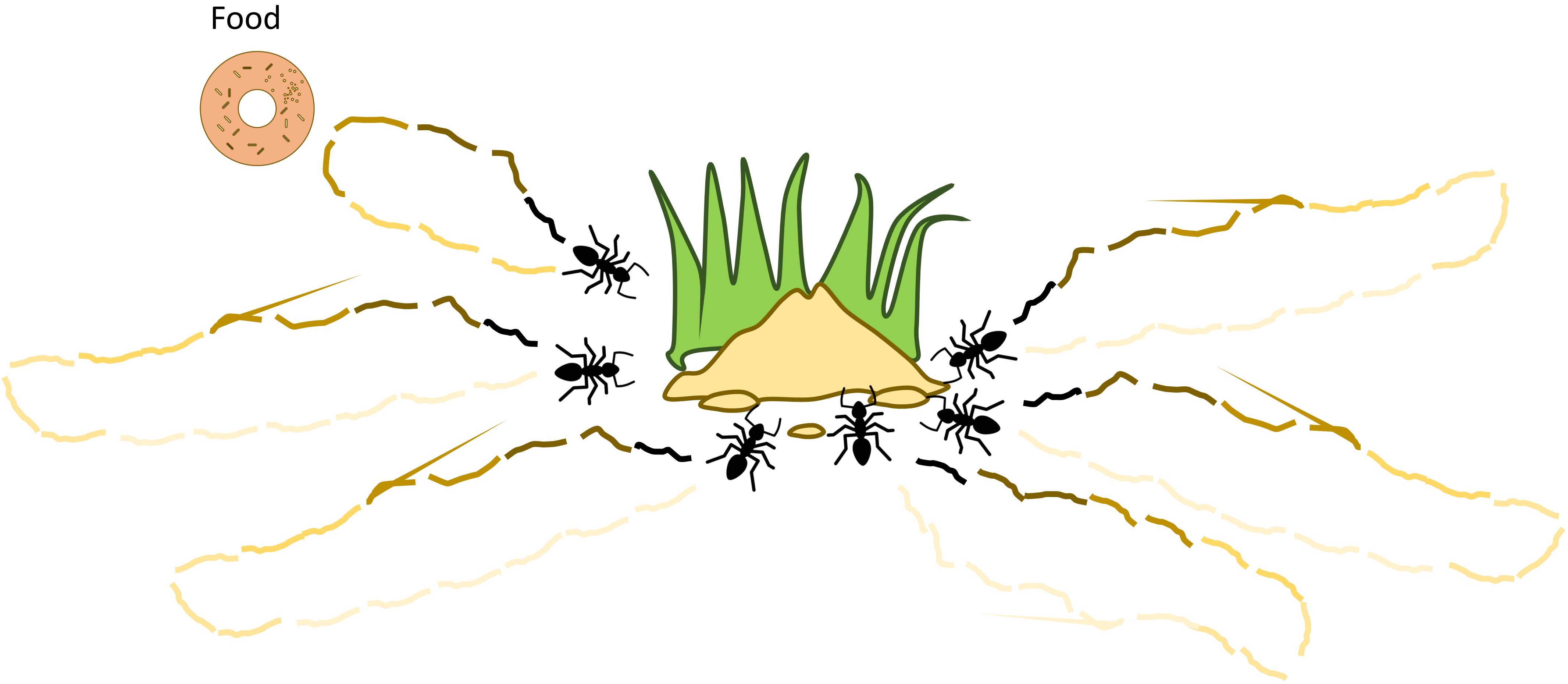




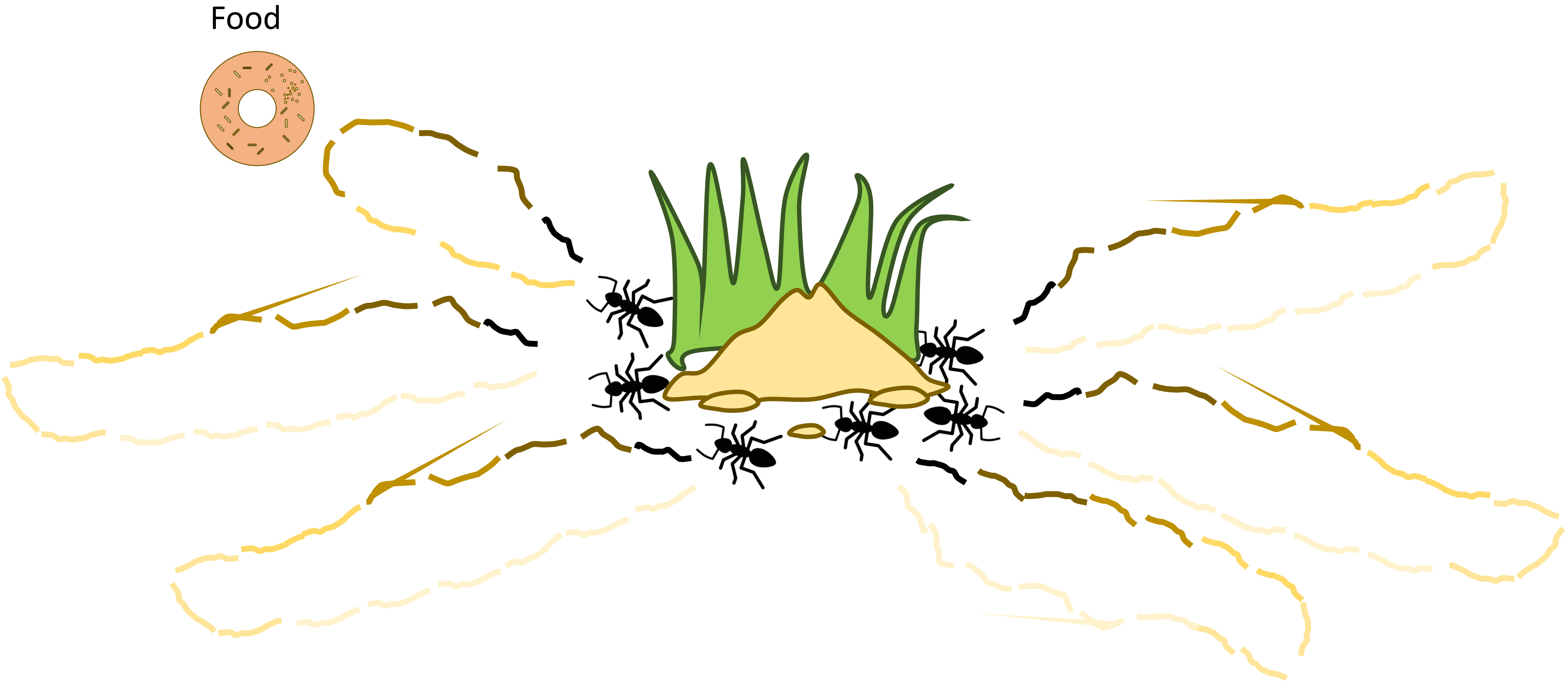
FOOD FORAGING BEHAVIOR



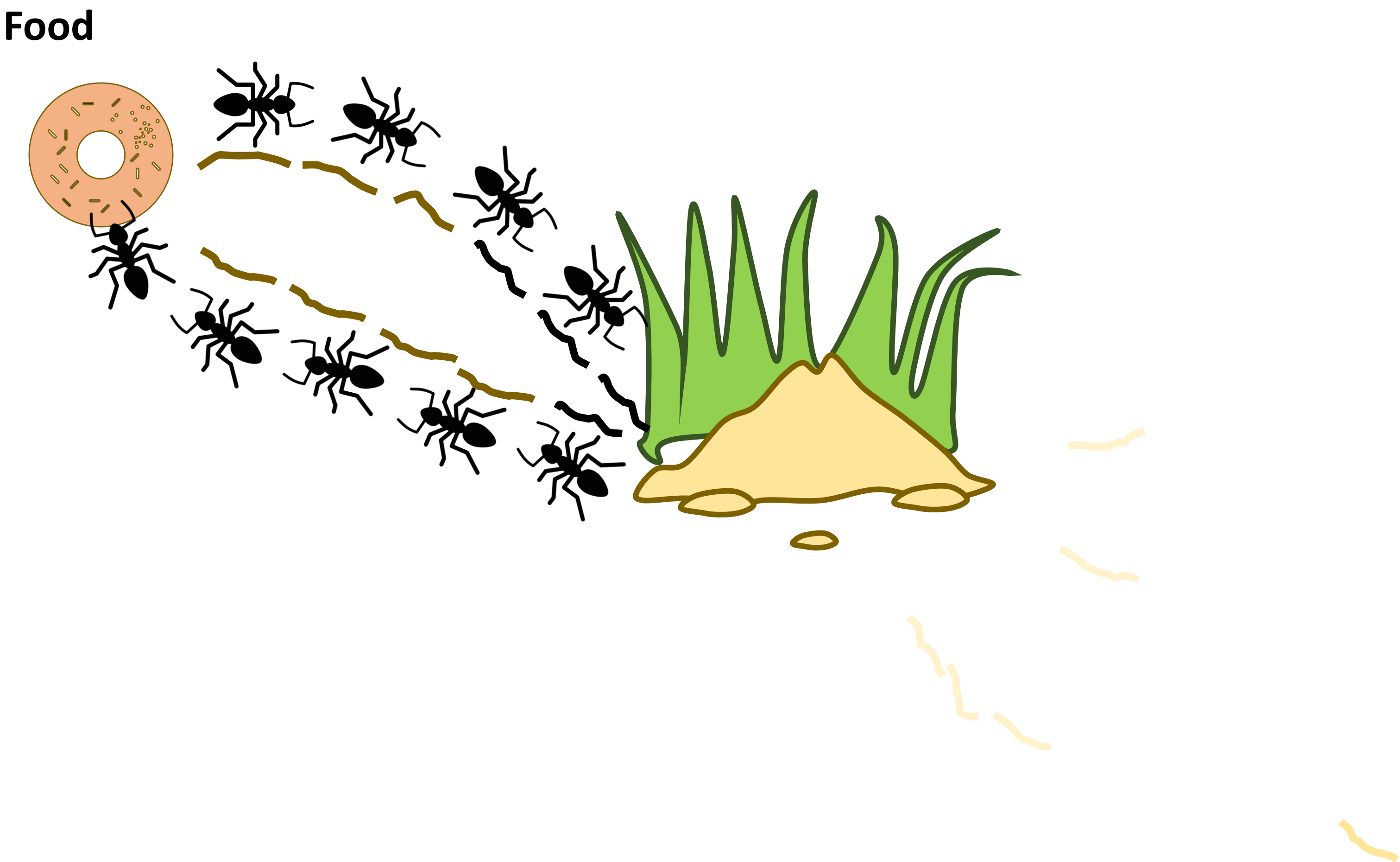
FOOD FORAGING BEHAVIOR



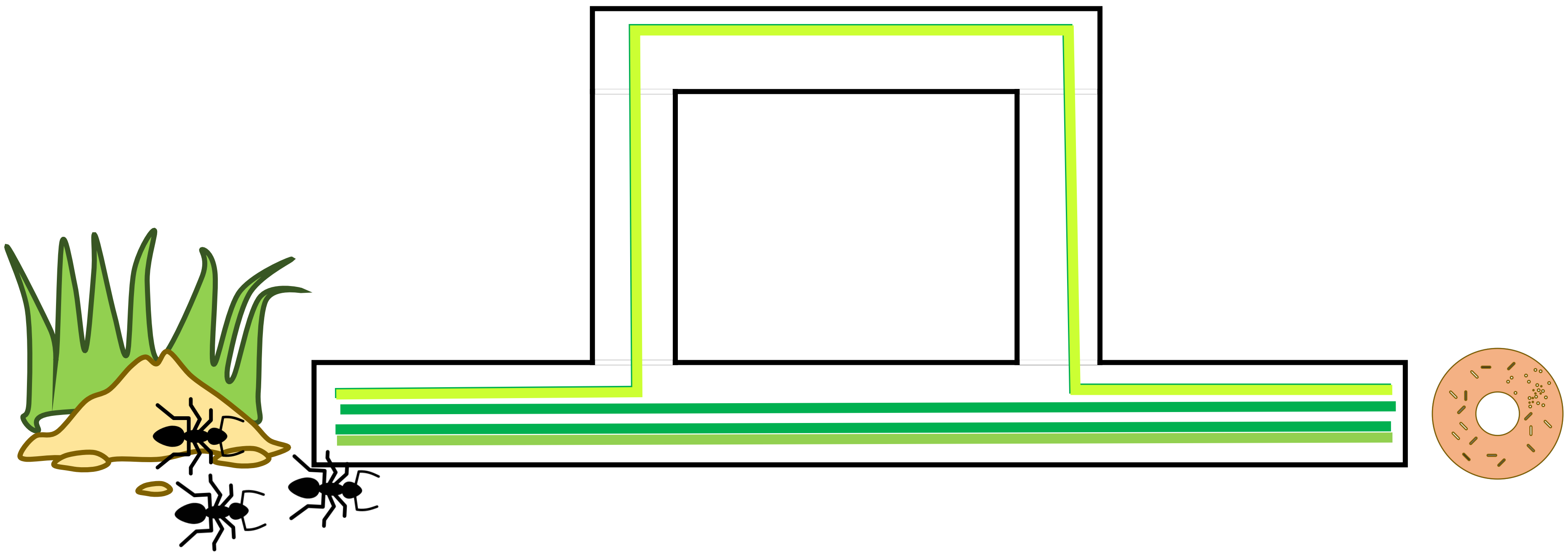
FOOD FORAGING BEHAVIOR



FOOD FORAGING BEHAVIOR



PHEROMONE CONCENTRATION AS A SHORTCUT

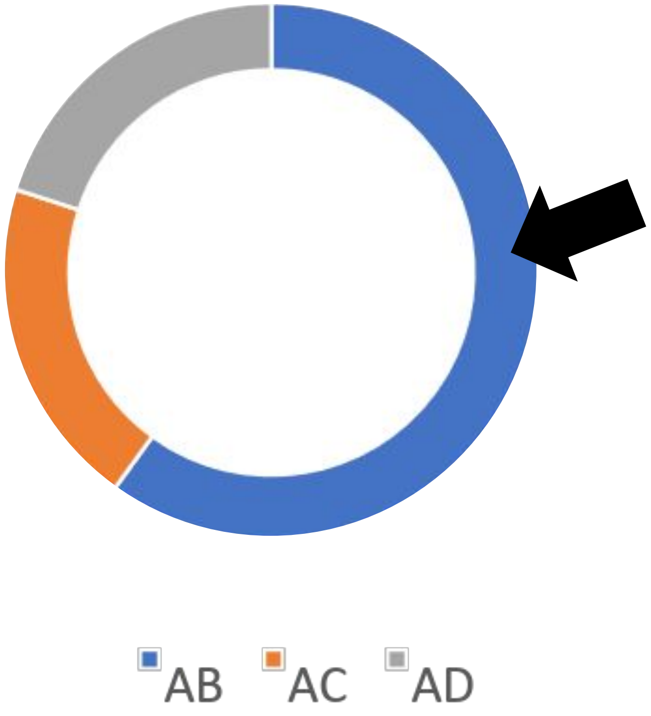


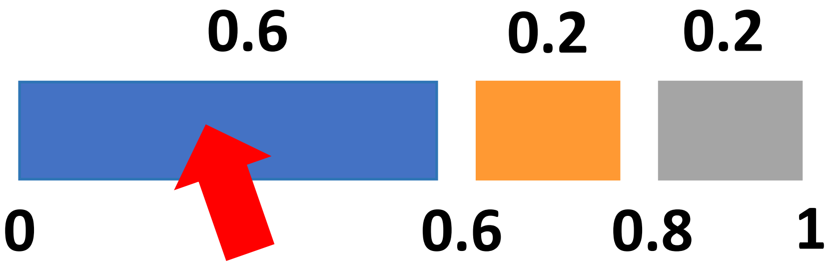
ROULLETE WHEEL TECHNIQUE

[AB,AC,AD]

[0.6,0.2,0.2]

AB: 0.6
AC: 0.2
AD: 0.2





count = 0

R = 0.34

If R < 0.6: count = 0

If R < 0.8: count = 1

If R > 0.8: count = 2

Index [0]

[AB,AC,AD]

[0.6,0.2,0.2]

PHEROMONE UPDATE

$$\Delta\tau^{k}_{x,y} = \{ 1/L_k \}$$

K = ant

x,**y** = edge from **x** to **y**

L_k = length of the path

Current level of pheromone deposited

$$\Delta\tau_{x,y} = (1 - \boxed{p})\boxed{\tau_{x,y}} + \boxed{\sum_{k=1}^{m} \Delta\tau^{k}_{x,y}}$$

Evap. constant

Pheromone to the added

PHEROMONE UPDATE

m = n. of ants

Current level of pheromone deposited

$$\Delta\tau_{x,y} = (1 - \boxed{p})\boxed{\tau_{x,y}} + \boxed{\sum_{k=1}^{m} \Delta\tau_{x,y}^{k}}$$

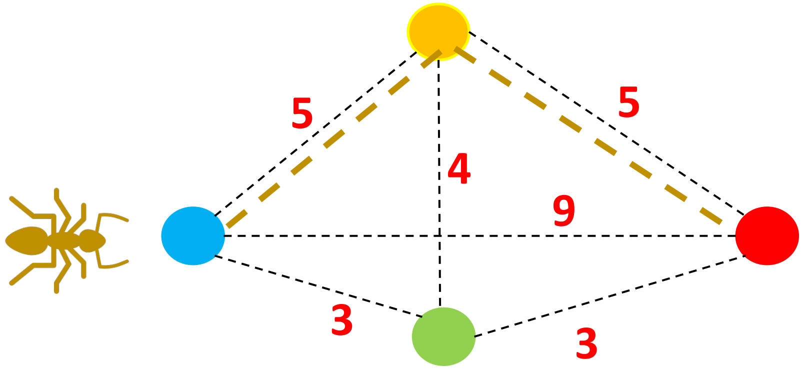
Evap. constant

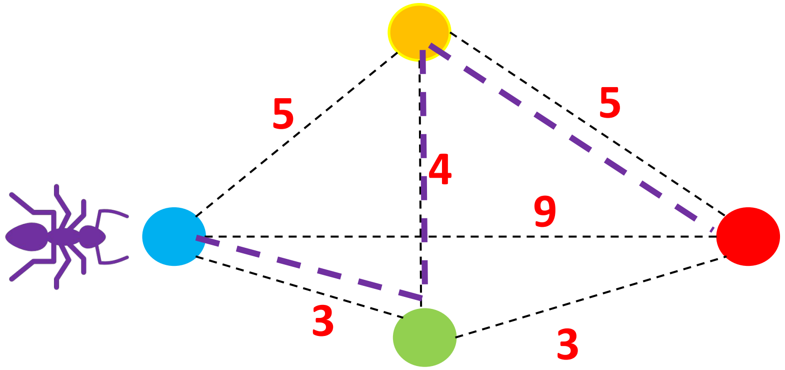
Pheromone to the added

$$\Delta\tau_{x,y}^{k} = \{ 1/L_k \}$$

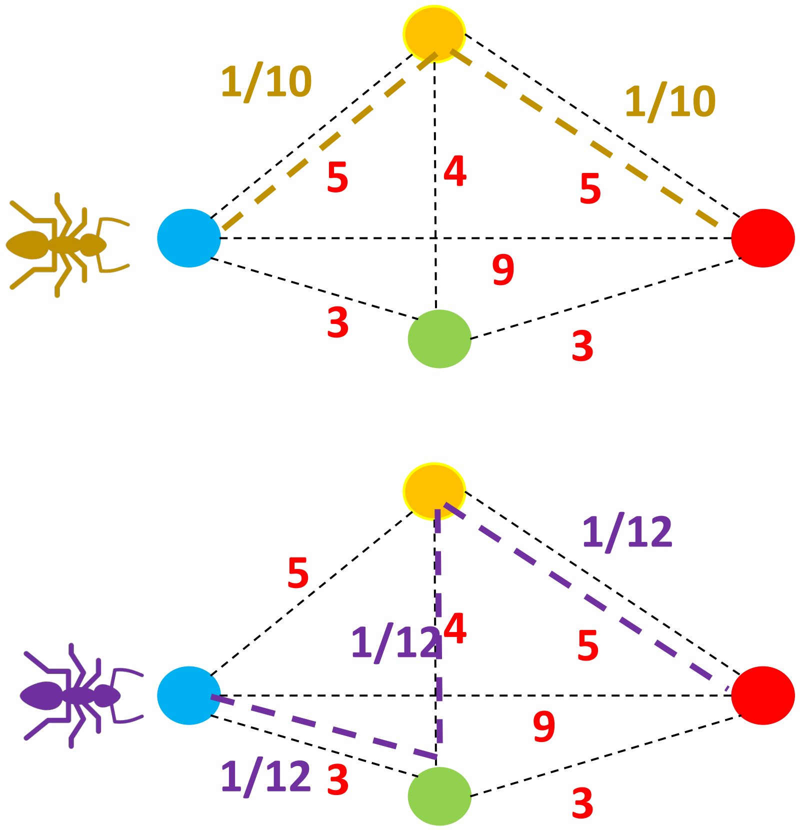
In turn, the amount of pheromone deposited by a certain ant **k** on the path **x,y** is given by **1/Lk**, that is, **1/length of the path**

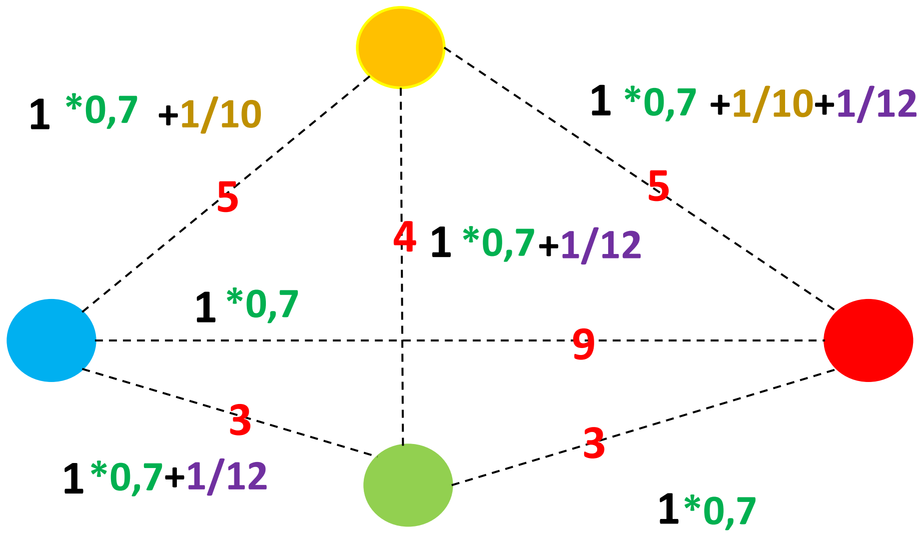
Const. Evaporation ρ: **0.3**





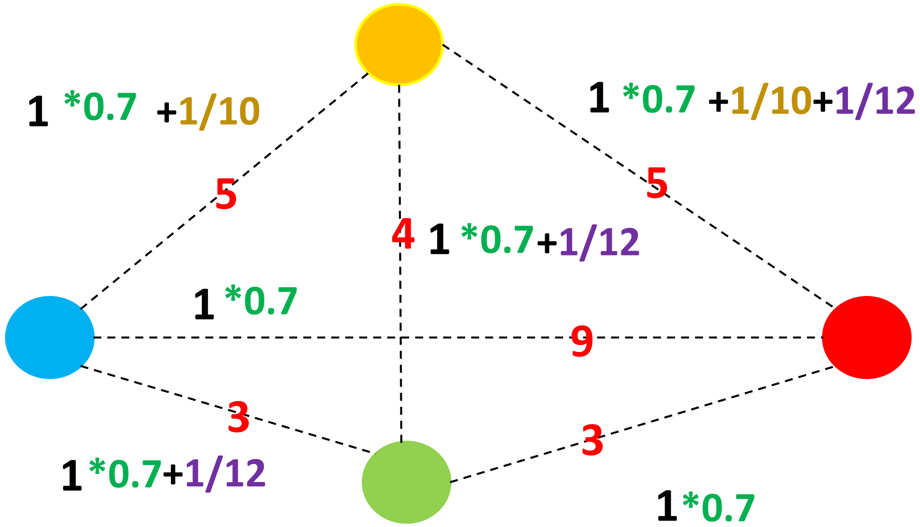
PHEROMONE UPDATE

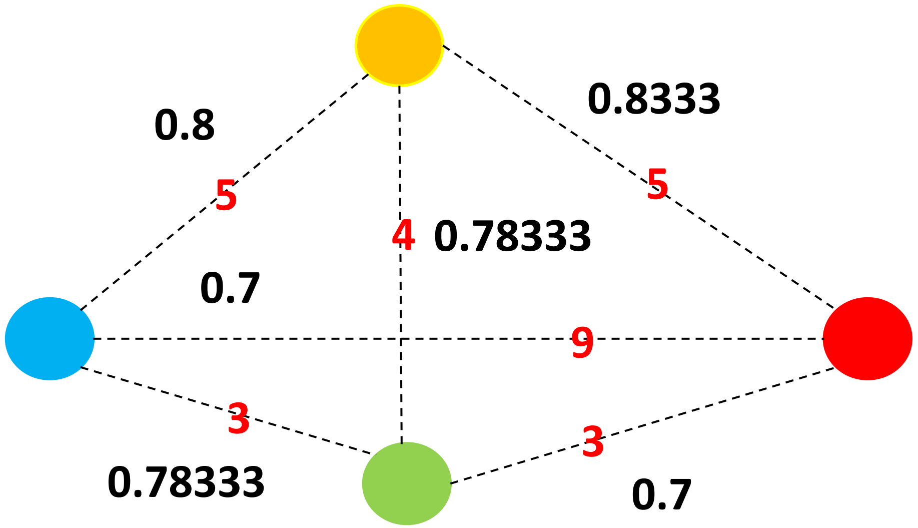




$$\tau_{x,y} = (1-\rho)\,\tau_{x,y} + \tau_{x,y(1)} + \tau_{x,y(2)}$$

PHEROMONE UPDATE

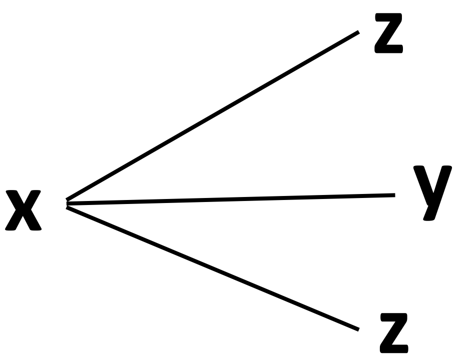




PROBABILITY OF EDGE SELECTION

Probability to choose the edge from point **x** to point **y**

$$p^{k}_{x,y} = \frac{\left(\tau^{\alpha}_{xy}\right)\left(\rho^{\beta}_{xy}\right)}{\sum z\ \varepsilon\ x\ allowed\ \left(\tau^{\alpha}_{xz}\right)\left(\rho^{\beta}_{xz}\right)}$$



$\left(\tau^{\alpha}_{xy}\right)$ = pheromone deposit

$\left(\rho^{\beta}_{xy}\right)$ = path quality $\Big\{$ (1/path length)

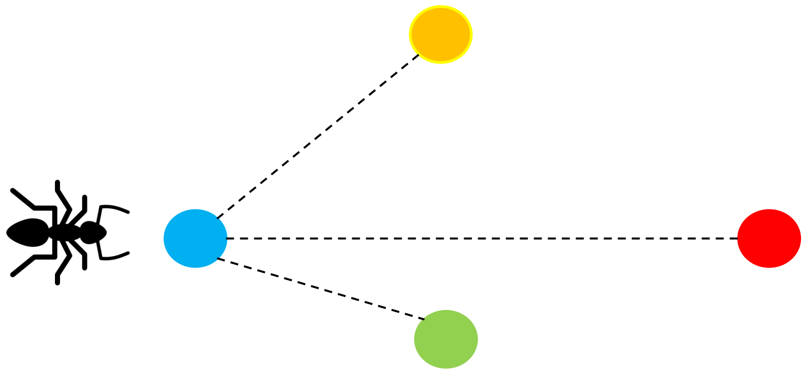
α and **β** are values greater than 0 that are used to control the influence of the pheromone deposit and the path distance. Most often they are set at 1.

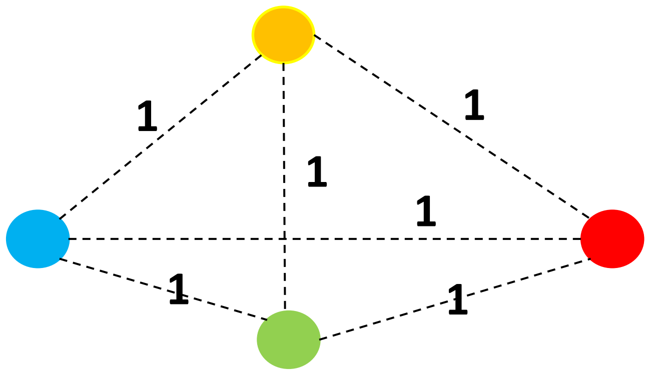
PROBABILITY OF EDGE SELECTION

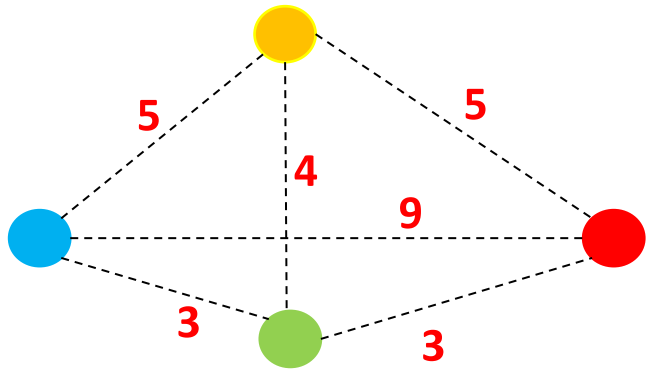
Probability to choose the edge from point **x** to point **y** ($p_{x,y}^{k}$)

$$p_{x,y}^{k} = \frac{\left(\tau_{xy}^{\alpha}\right)\left(\rho_{xy}^{\beta}\right)}{\sum z \; \varepsilon \; x \; allowed \; \left(\tau_{xz}^{\alpha}\right)\left(\rho_{xz}^{\beta}\right)}$$

$\left(\tau_{xy}^{\alpha}\right)$ = pheromone deposit $\left(\rho_{xy}^{\beta}\right)$ = path quality



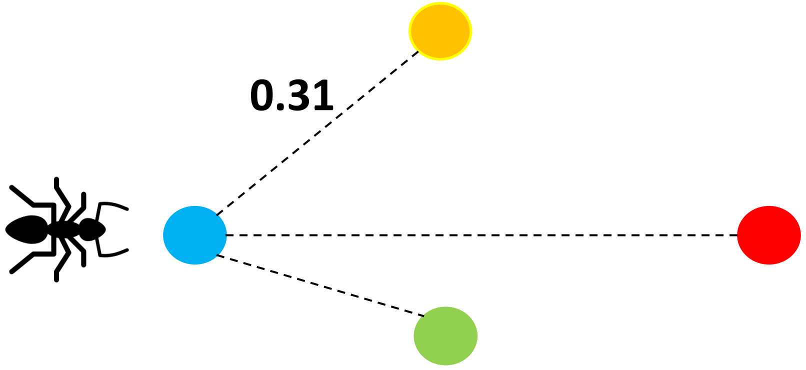




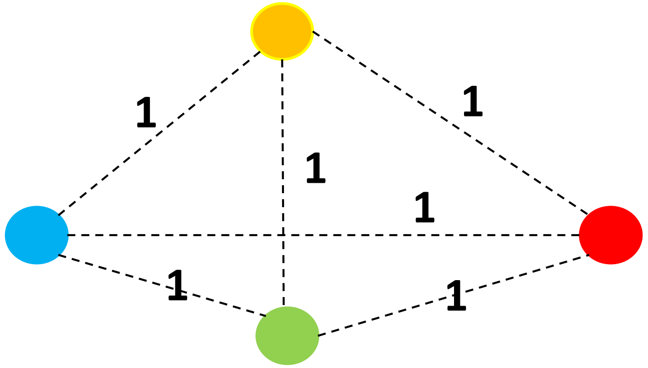
PROBABILITY OF EDGE SELECTION

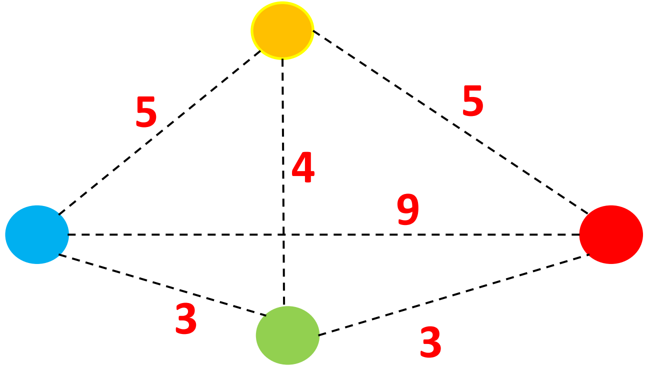
$$p^{k}_{x,y}= \frac{\left(\tau^{\alpha}_{xy}\right)\left(\rho^{\beta}_{xy}\right)}{\sum z\ \varepsilon\ x\ allowed\ \left(\tau^{\alpha}_{xz}\right)\left(\rho^{\beta}_{xz}\right)}$$

$\left(\tau^{\alpha}_{xy}\right)$ = pheromone deposit $\left(\rho^{\beta}_{xy}\right)$ = path quality



$$p^{k}_{x,y}= \frac{(1)\ \mathrm{X}\ (1/5)}{(1)\ \mathrm{X}\ (1/5)\ +\ (1)\ \mathrm{X}\ (1/9)\ +\ (1)\ \mathrm{X}\ (1/3)}$$ 0.31



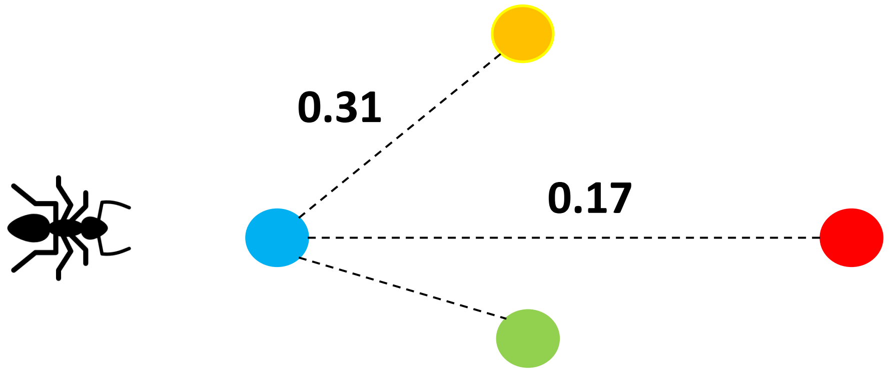


PROBABILITY OF EDGE SELECTION

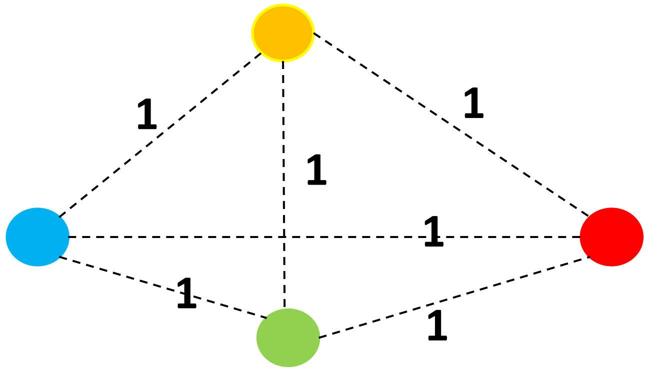
$$p^{k}_{x,y} = \frac{\left(\tau^{\alpha}_{xy}\right)\left(\rho^{\beta}_{xy}\right)}{\sum z \,\varepsilon\, x \; allowed \left(\tau^{\alpha}_{xz}\right)\left(\rho^{\beta}_{xz}\right)}$$

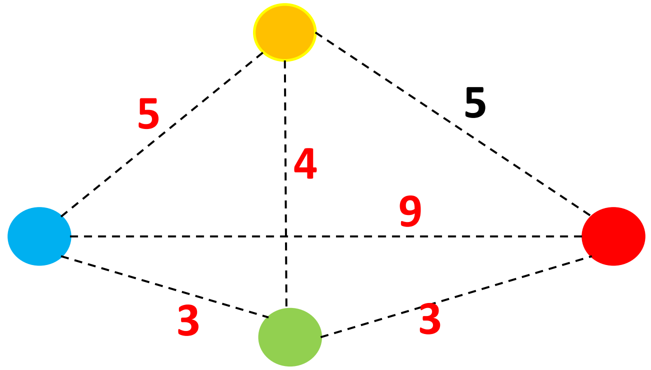
$\left(\tau^{\alpha}_{xy}\right)$ = pheromone deposit

$\left(\rho^{\beta}_{xy}\right)$ = path quality



$$p^{k}_{x,y} = \frac{(1) \text{ X } (1/9)}{(1) \text{ X } (1/5) + (1) \text{ X } (1/9) + (1) \text{ X } (1/3)} \quad 0.17$$

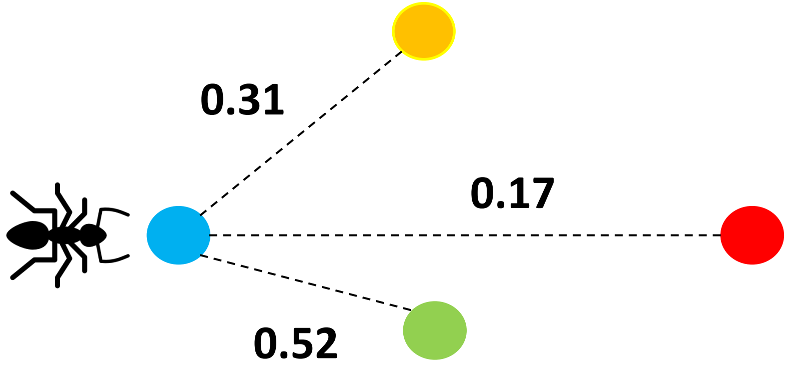




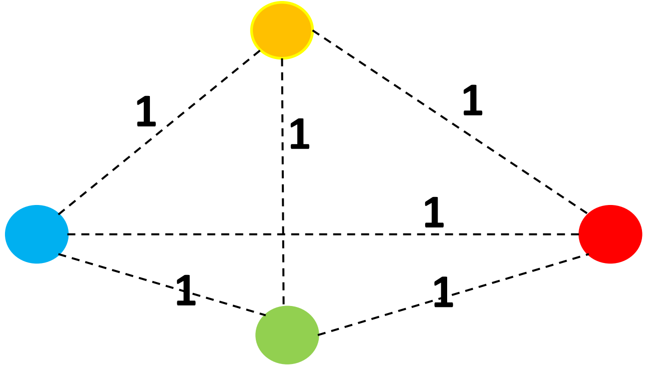
PROBABILITY OF EDGE SELECTION

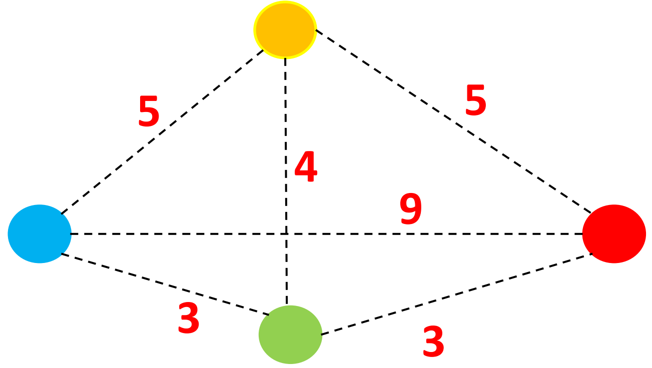
$$p_{x,y}^{k} = \frac{\left(\tau_{xy}^{\alpha}\right)\left(\rho_{xy}^{\beta}\right)}{\sum z \,\varepsilon\, x \; allowed \left(\tau_{xz}^{\alpha}\right)\left(\rho_{xz}^{\beta}\right)}$$

$\left(\tau_{xy}^{\alpha}\right)$ = pheromone deposit $\left(\rho_{xy}^{\beta}\right)$ = path quality



$$p_{x,y}^{k} = \frac{(1) \text{ X } (1/3)}{(1) \text{ X } (1/5) + (1) \text{ X } (1/9) + (1) \text{ X } (1/3)} \quad 0.52$$





Edges

AB: 1; **1**; (BD, BC)

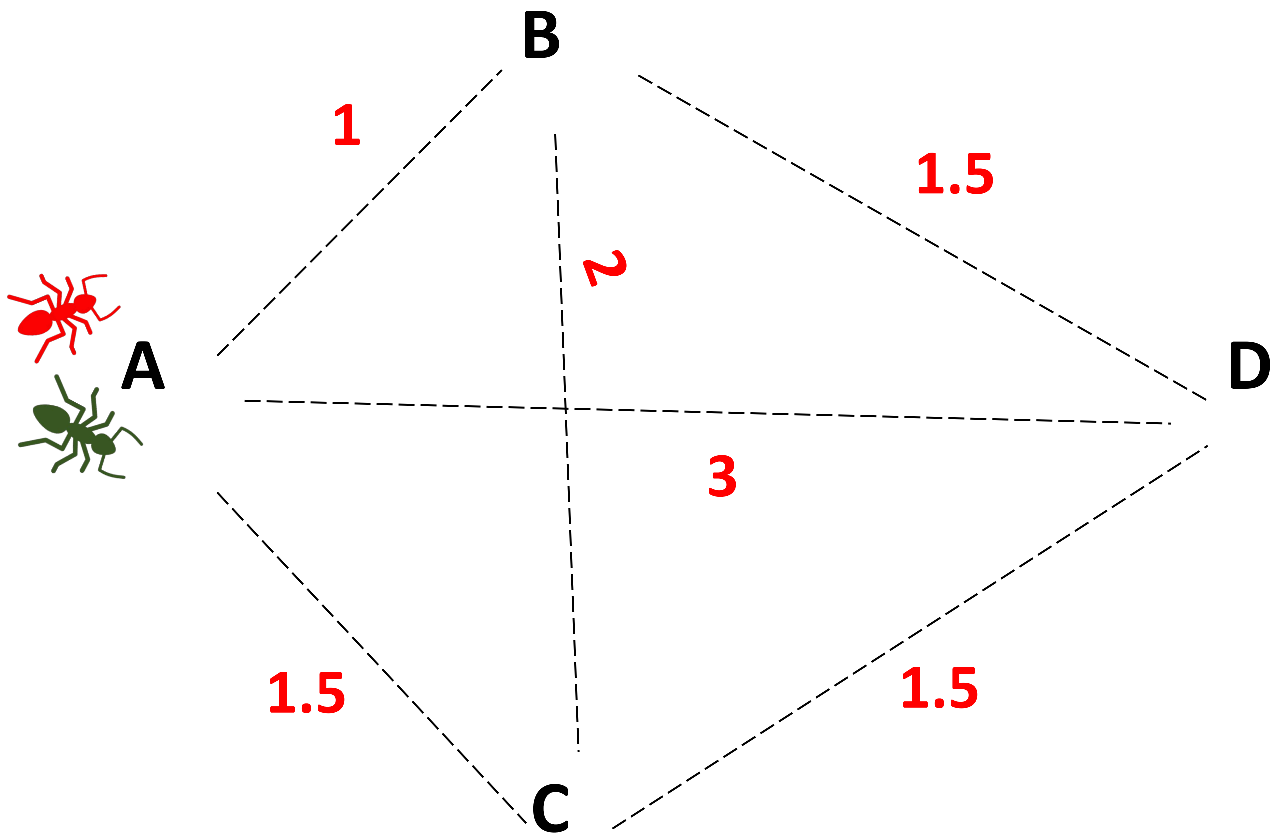
AD: 3; **1**; (---)

AC: 1.5; **1**; (CD)

BD: 1.5; **1**; (AB, CD)

CD: 1.5; **1**; (---)

BC: 2; **1**; (AC, AB, BD, CD)



Edges

AB: 1; **1**; (BD, BC)

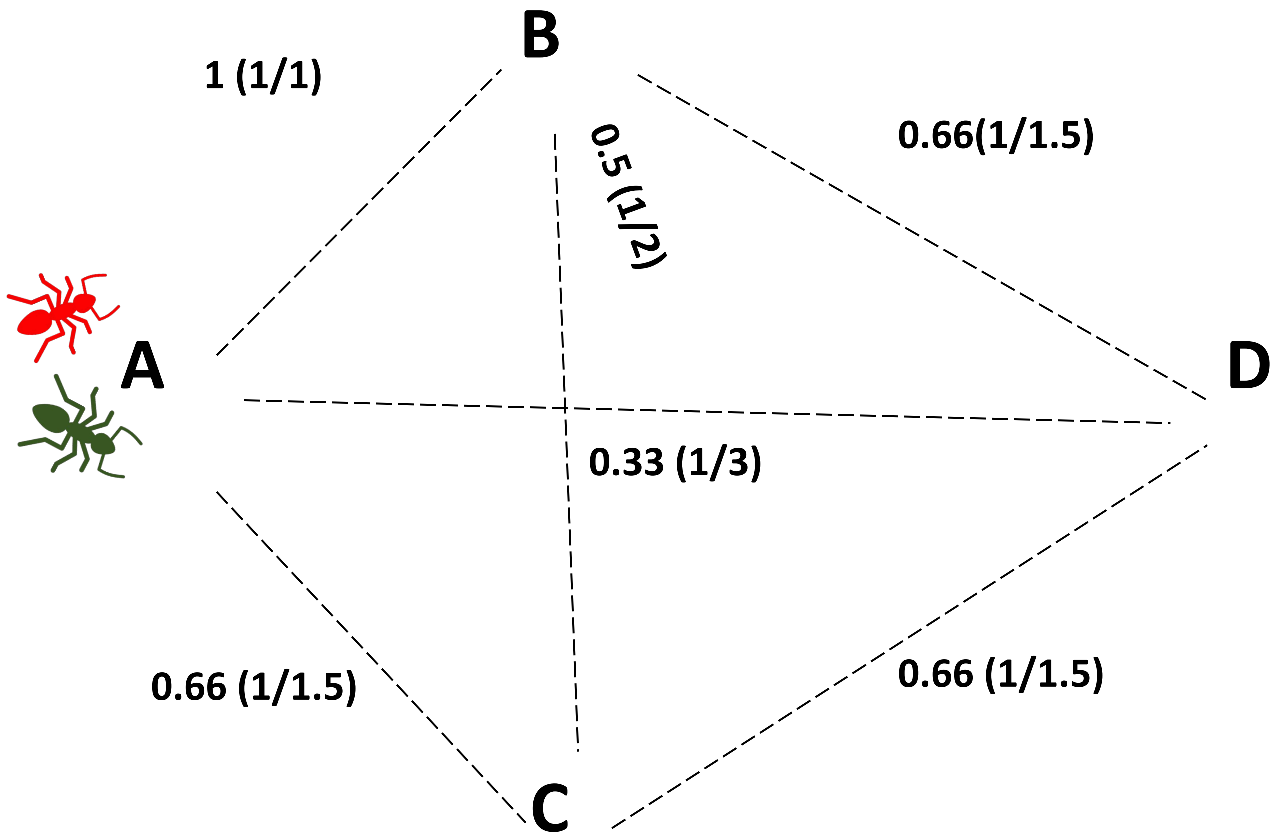
AD: 3; **1**; (---)

AC: 1.5; **1**; (CD)

BD: 1.5; **1**; (AB, CD)

CD: 1.5; **1**; (---)

BC: 2; **1**; (AC, AB, BD, CD)



Edges

AB: 1; **1**; (BD, BC)

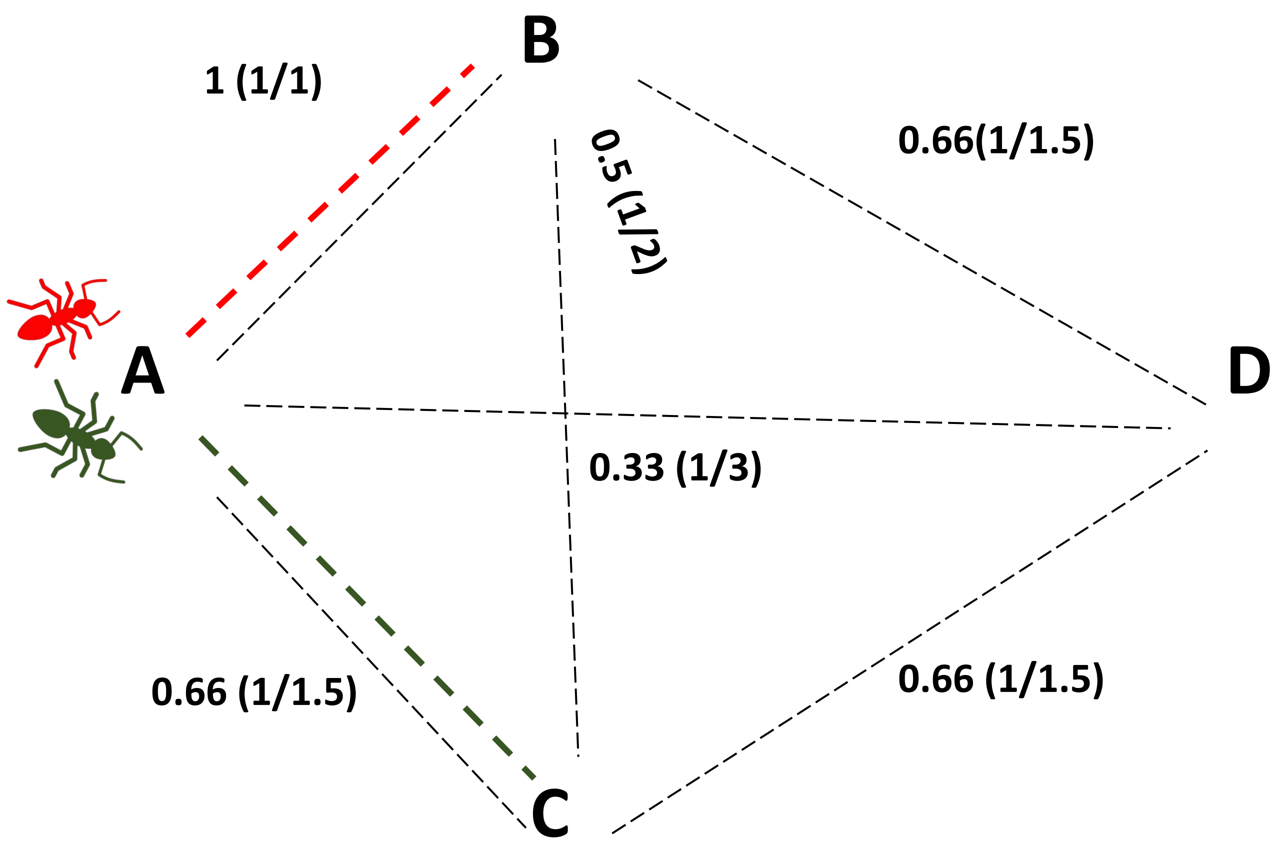
AD: 3; **1**; (---)

AC: 1.5; **1**; (CD)

BD: 1.5; **1**; (AB, CD)

CD: 1.5; **1**; (---)

BC: 2; **1**; (AC, AB, BD, CD)



Edges

AB: 1; **1**; (BD, BC)

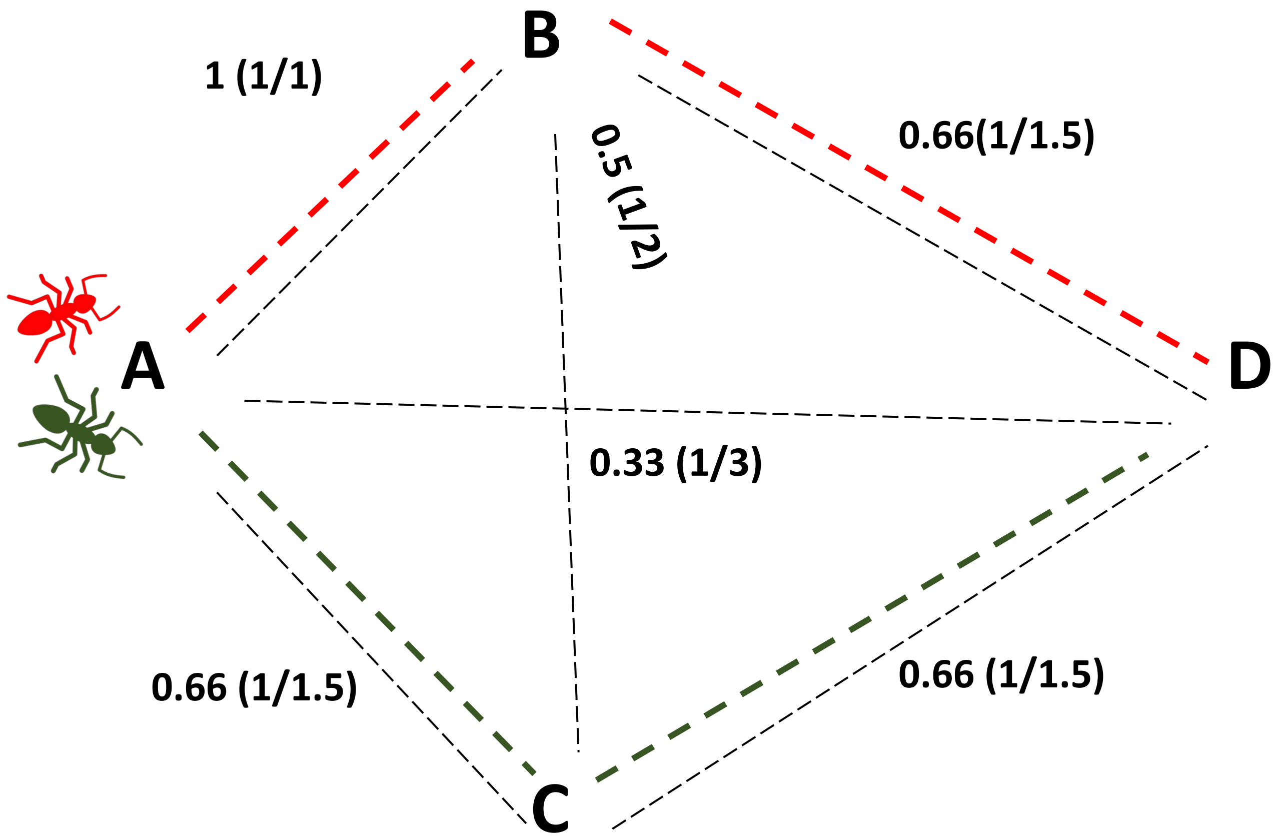
AD: 3; **1**; (---)

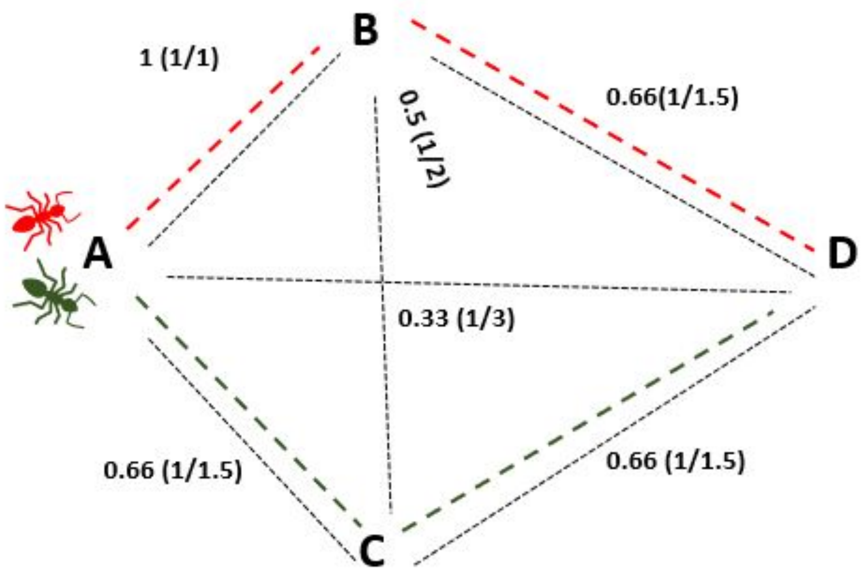
AC: 1.5; **1**; (CD)

BD: 1.5; **1**; (AB, CD)

CD: 1.5; **1**; (---)

BC: 2; **1**; (AC, AB, BD, CD)





2 Ants
AB,BD
Path length: 2.5
Pheromone deposited: 1/2.5 (0.4)

AC, CD
Path length : 3
Pheromone deposited : 1/3 (0.33)

Edges
AB: 1; **1**; (BD, BC)
AD: 3; **1**; (---)
AC: 1.5; **1**; (CD)
BD: 1.5; **1**; (AB, CD)
CD: 1.5; **1**; (---)
BC: 2; **1**; (AC, AB, BD, CD)

Edges
AB: 1; **0.7*1+0.4**; (BD, BC)
AD: 3; **0.7*1**; (---)
AC: 1.5; **0.7*1+0.33**; (CD)
BD: 1.5; **0.7*1+0.4** (AB, CD)
CD: 1.5; **0.7*1+0.33**; (---)
BC: 2; **0.7*1**; (AC, AB, BD, CD)

Edges
AB: 1; **1.1**; (BD, BC)
AD: 3; **0.7**; (---)
AC: 1.5; **1.033**; (CD)
BD: 1.5; **1.1**; (AB, CD)
CD: 1.5; **1.033**; (---)
BC: 2; **0.7**; (AC, AB, BD, CD)

Edges

AB: 1; **1.1**; (BD, BC)

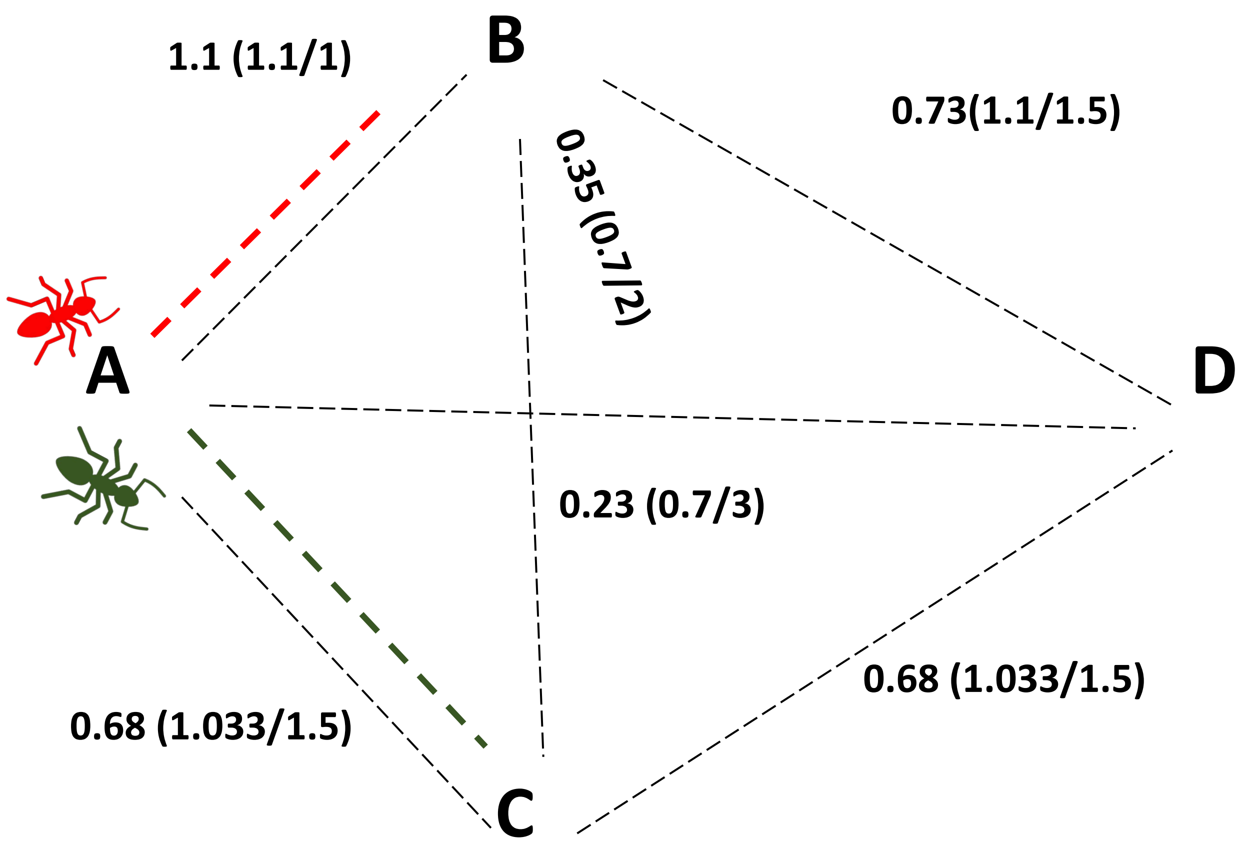
AD: 3; **0.7**; (---)

AC: 1.5; **1.033**; (CD)

BD: 1.5; **1.1**; (AB, CD)

CD: 1.5; **1.033**; (---)

BC: 2; **0.7**; (AC, AB, BD, CD)



Edges

AB: 1; **1.1**; (BD, BC)

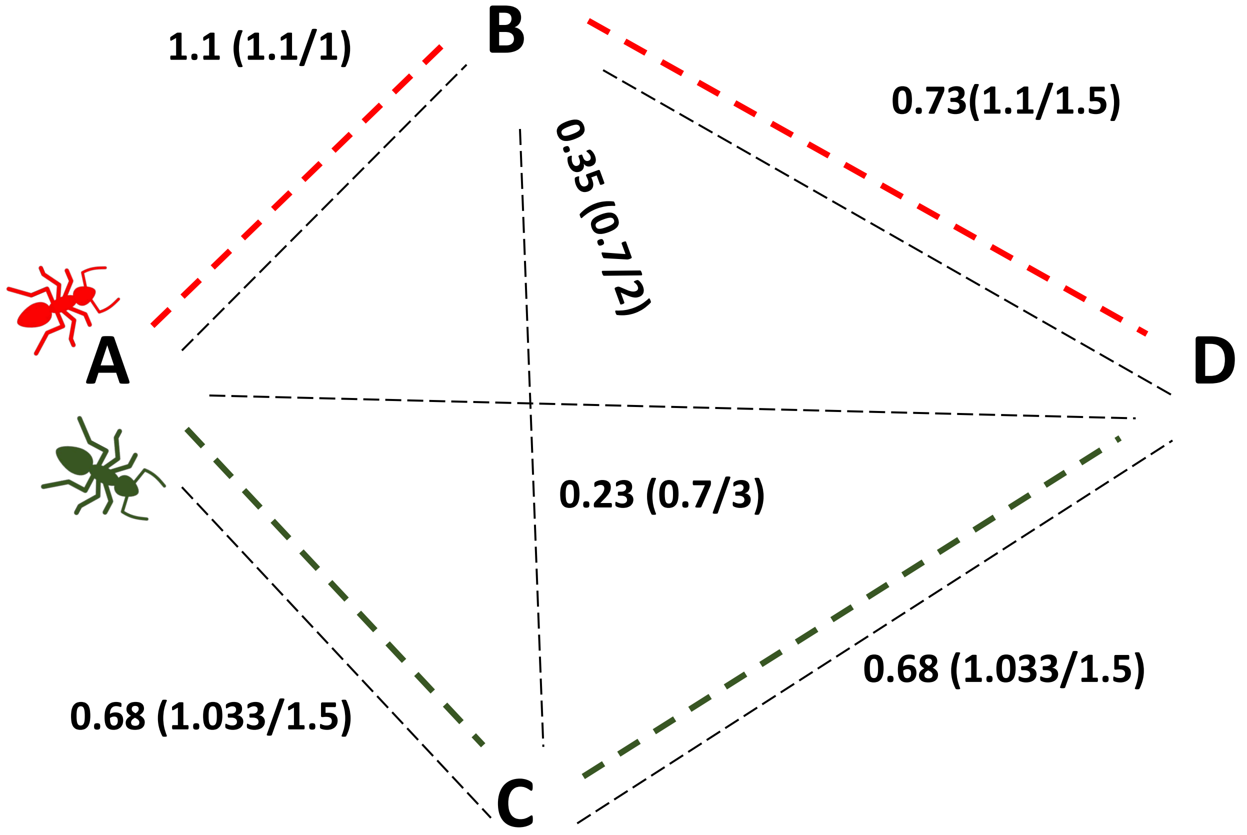
AD: 3; **0.7**; (---)

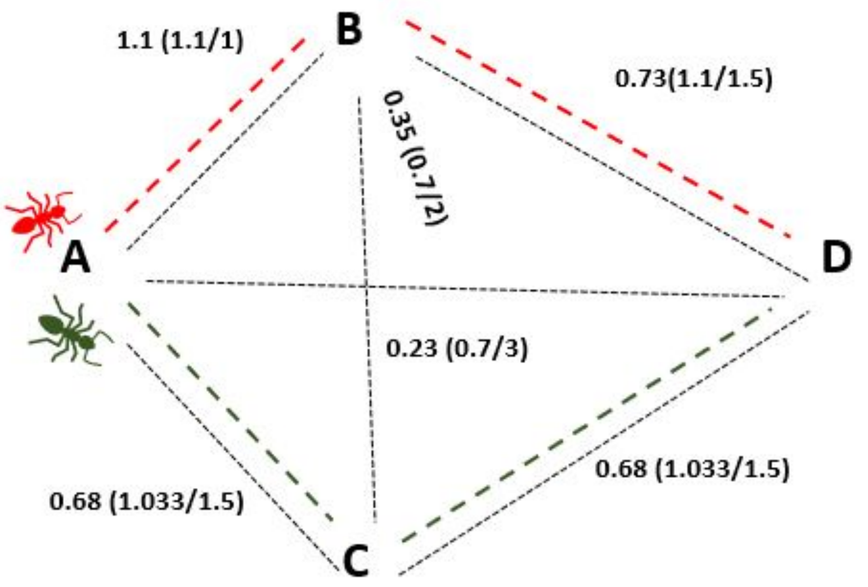
AC: 1.5; **1.033**; (CD)

BD: 1.5; **1.1**; (AB, CD)

CD: 1.5; **1.033**; (---)

BC: 2; **0.7**; (AC, AB, BD, CD)





2 Ants

AB,BD

Path length: 2.5

Pheromone deposited: 1/2.5 (0.4)

AB, CD

Path length : 3

Pheromone deposited : 1/3 (0.33)

Edges

AB: 1; **1.1**; (BD, BC)

AD: 3; **0.7**; (---)

AC: 1.5; **1.033**; (CD)

BD: 1.5; **1.1**; (AB, CD)

CD: 1.5; **1.033**; (---)

BC: 2; **0.7**; (AC, AB, BD, CD)

Edges

AB: 1; **0.7*1.1+0.4**; (BD, BC)

AD: 3; **0.7*0.7**; (---)

AC: 1.5; **0.7*1.033+0.33**; (CD)

BD: 1.5; **0.7*1.1+0.4** (AB, CD)

CD: 1.5; **0.7*1.033+0.33**; (---)

BC: 2; **0.7*0.7**; (AC, AB, BD, CD)

Edges

AB: 1; **1.17**; (BD, BC)

AD: 3; **0.49**; (---)

AC: 1.5; **1.05**; (CD)

BD: 1.5; **1.17** (AB, CD)

CD: 1.5; **1.05**; (---)

BC: 2; **0.49**; (AC, AB, BD, CD)

Edges

AB: 1; **1.17**; (BD, BC)

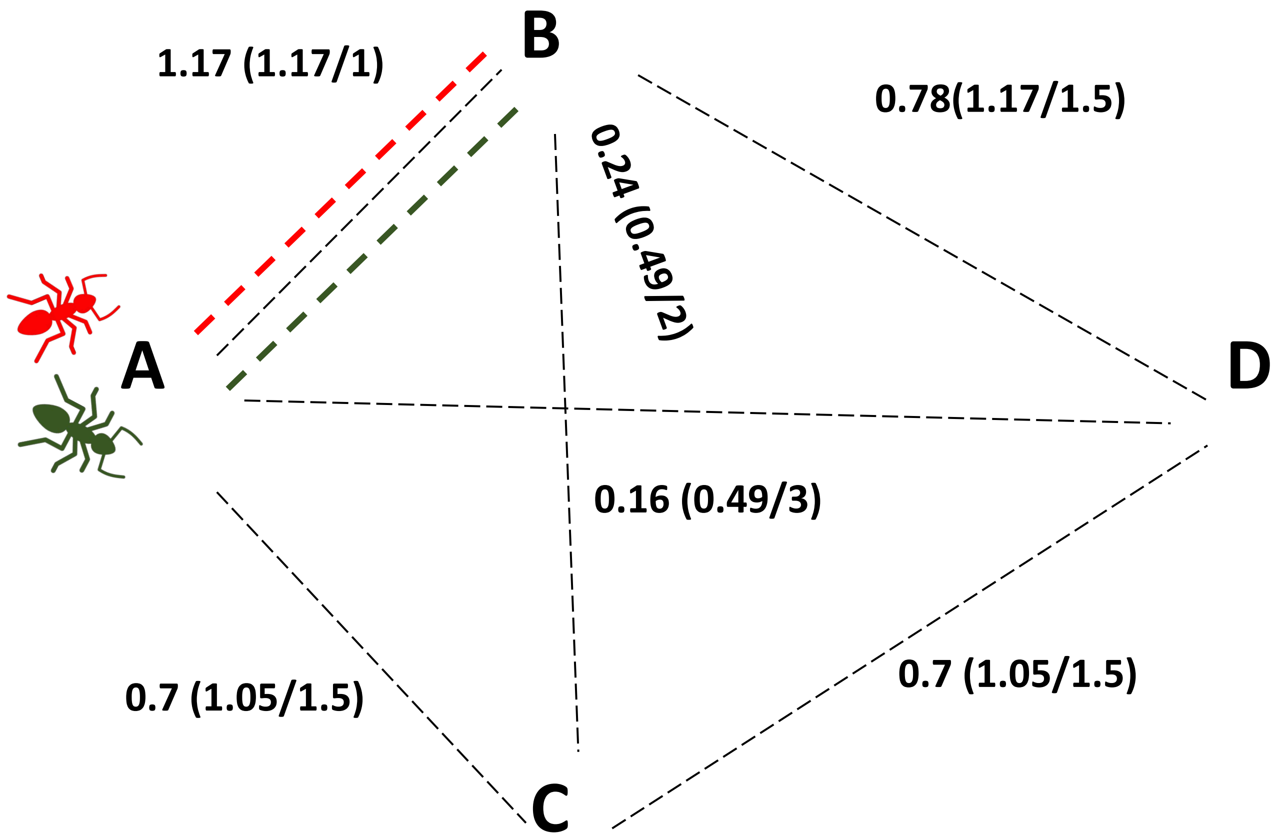
AD: 3; **0.49**; (---)

AC: 1.5; **1.05**; (CD)

BD: 1.5; **1.17** (AB, CD)

CD: 1.5; **1.05**; (---)

BC: 2; **0.49**; (AC, AB, BD, CD)



Edges

AB: 1; **1.17**; (BD, BC)

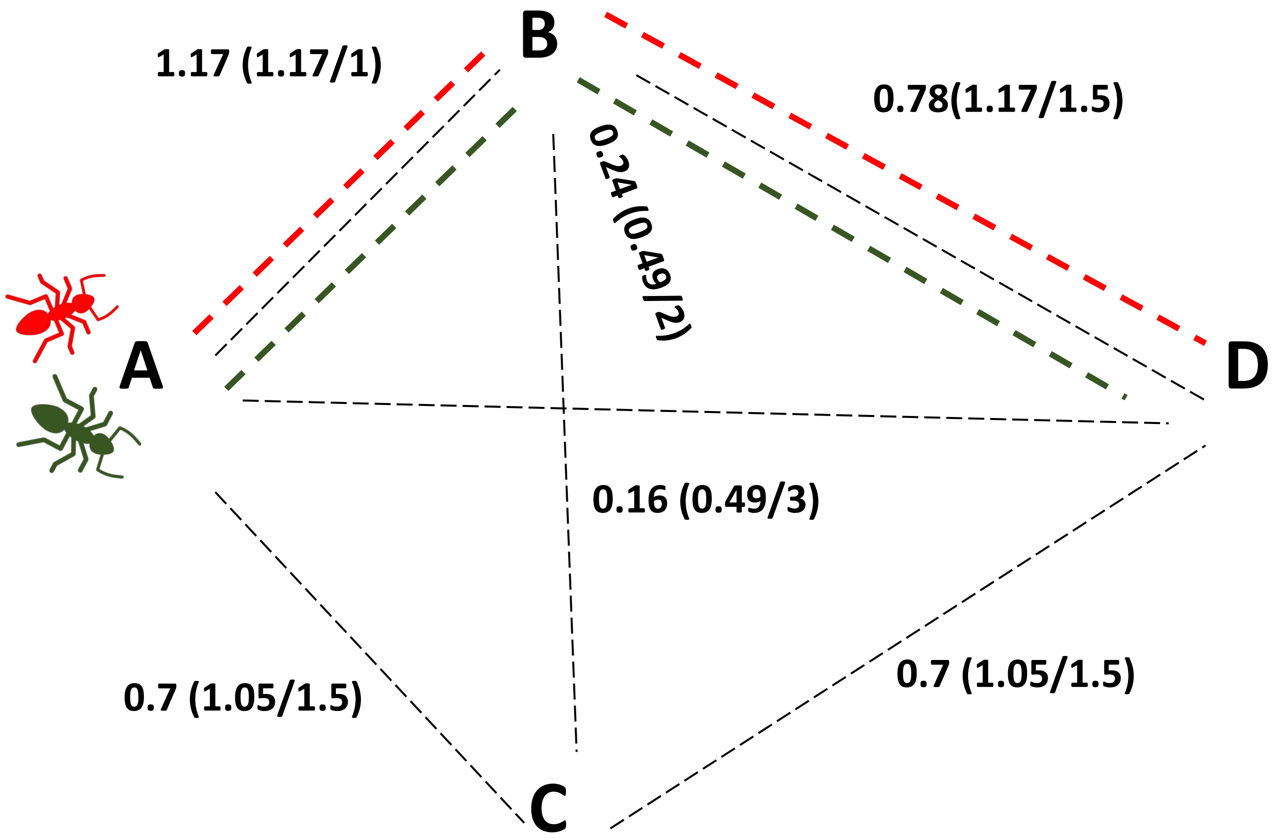
AD: 3; **0.49**; (---)

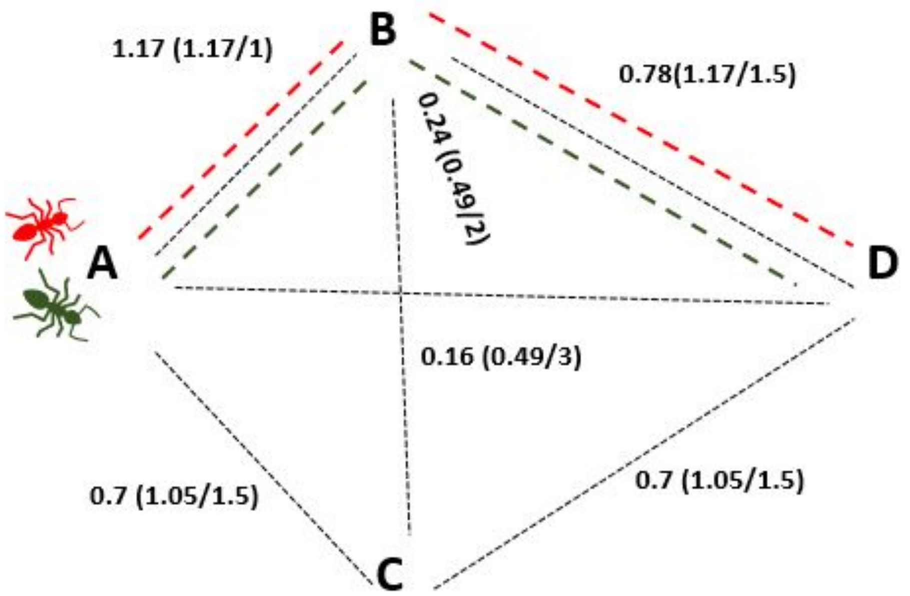
AC: 1.5; **1.05**; (CD)

BD: 1.5; **1.17** (AB, CD)

CD: 1.5; **1.05**; (---)

BC: 2; **0.49**; (AC, AB, BD, CD)





2 Ants
AB,BD
Path length: 2.5
Pheromone deposition: 1/2.5 (0.4)

AB, BD
Path length : 2.5
Pheromone deposition: 1/2.5 (0.4)

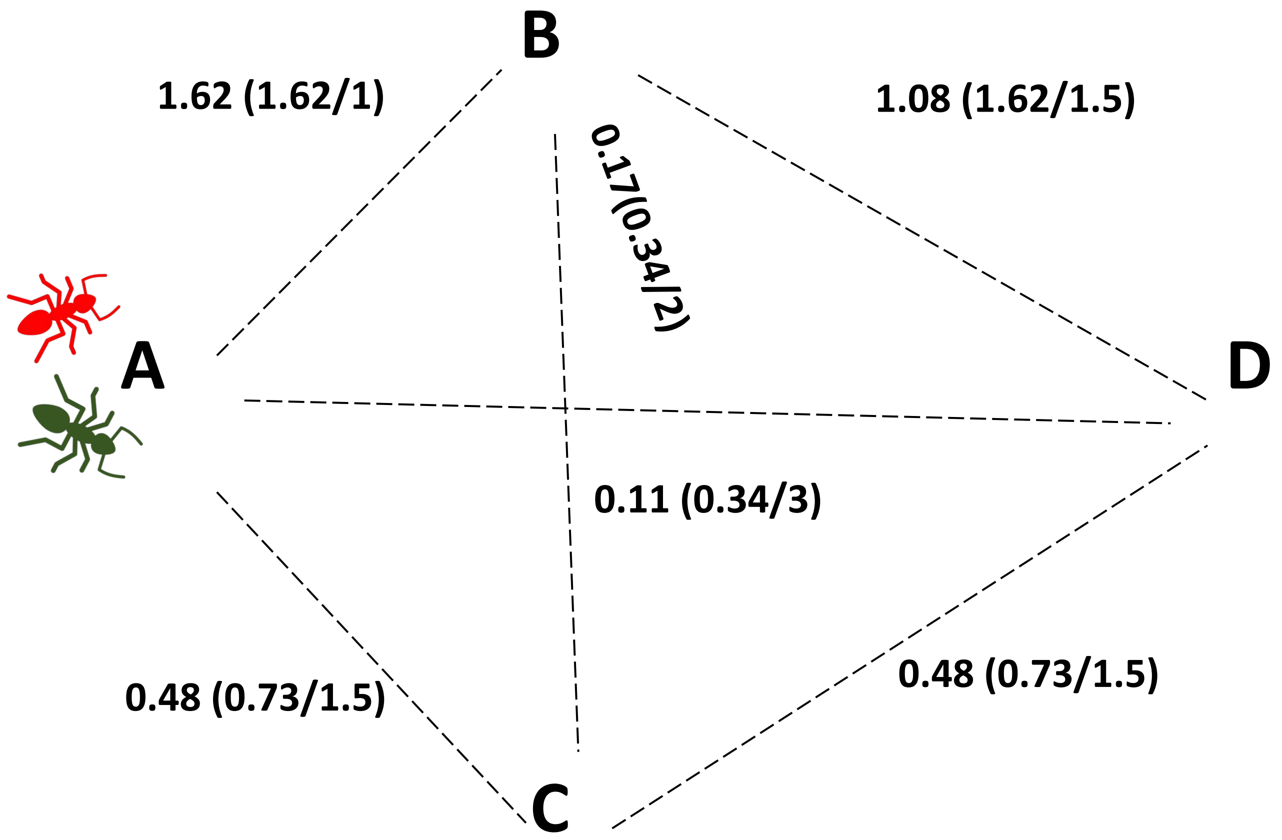
Edges
AB: 1; **1.17**; (BD, BC)
AD: 3; **0.49**; (---)
AC: 1.5; **1.05**; (CD)
BD: 1.5; **1.17** (AB, CD)
CD: 1.5; **1.05**; (---)
BC: 2; **0.49**; (AC, AB, BD, CD)

Edges
AB: 1; 0.7* **1.17**+0.4+0.4; (BD, BC)
AD: 3; 0.7* **0.49**; (---)
AC: 1.5; 0.7* **1.05**; (CD)
BD: 1.5; 0.7* **1.17**+0.4+0.4 (AB, CD)
CD: 1.5; 0.7* **1.05**; (---)
BC: 2; 0.7* **0.49**; (AC, AB, BD, CD)

Edges
AB: 1; **1.62**; (BD, BC)
AD: 3; **0.34**; (---)
AC: 1.5; **0.73**; (CD)
BD: 1.5; **1.62** (AB, CD)
CD: 1.5; **0.73**; (---)
BC: 2; **0.34**; (AC, AB, BD, CD)

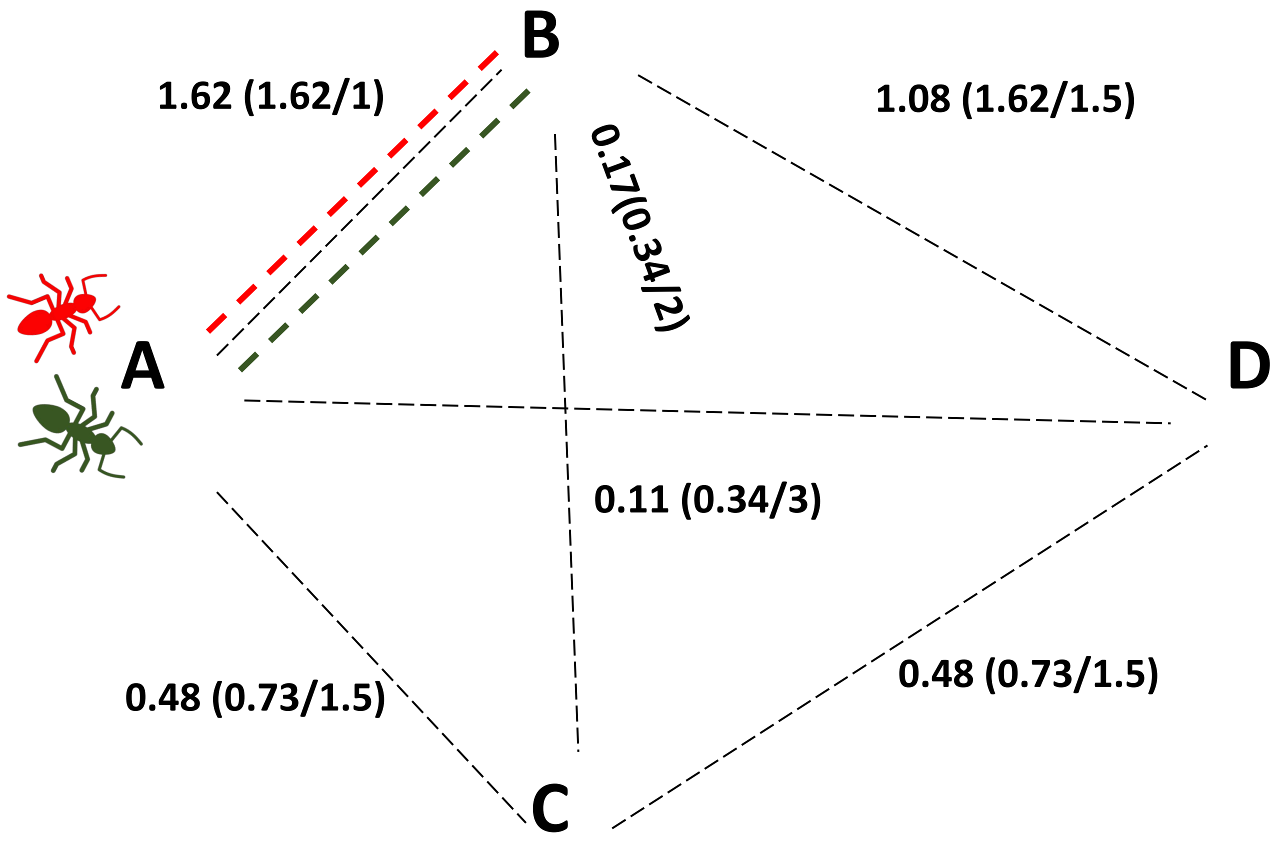
Edges

AB: 1; **1.62**; (BD, BC)
AD: 3; **0.34**; (---)
AC: 1.5; **0.73**; (CD)
BD: 1.5; **1.62** (AB, CD)
CD: 1.5; **0.73**; (---)
BC: 2; **0.34**; (AC, AB, BD, CD)



Edges

AB: 1; **1.62**; (BD, BC)
AD: 3; **0.34**; (---)
AC: 1.5; **0.73**; (CD)
BD: 1.5; **1.219** (AB, CD)
CD: 1.5; **1.535**; (---)
BC: 2; **0.34**; (AC, AB, BD, CD)



Edges

AB: 1; **1.62**; (BD, BC)

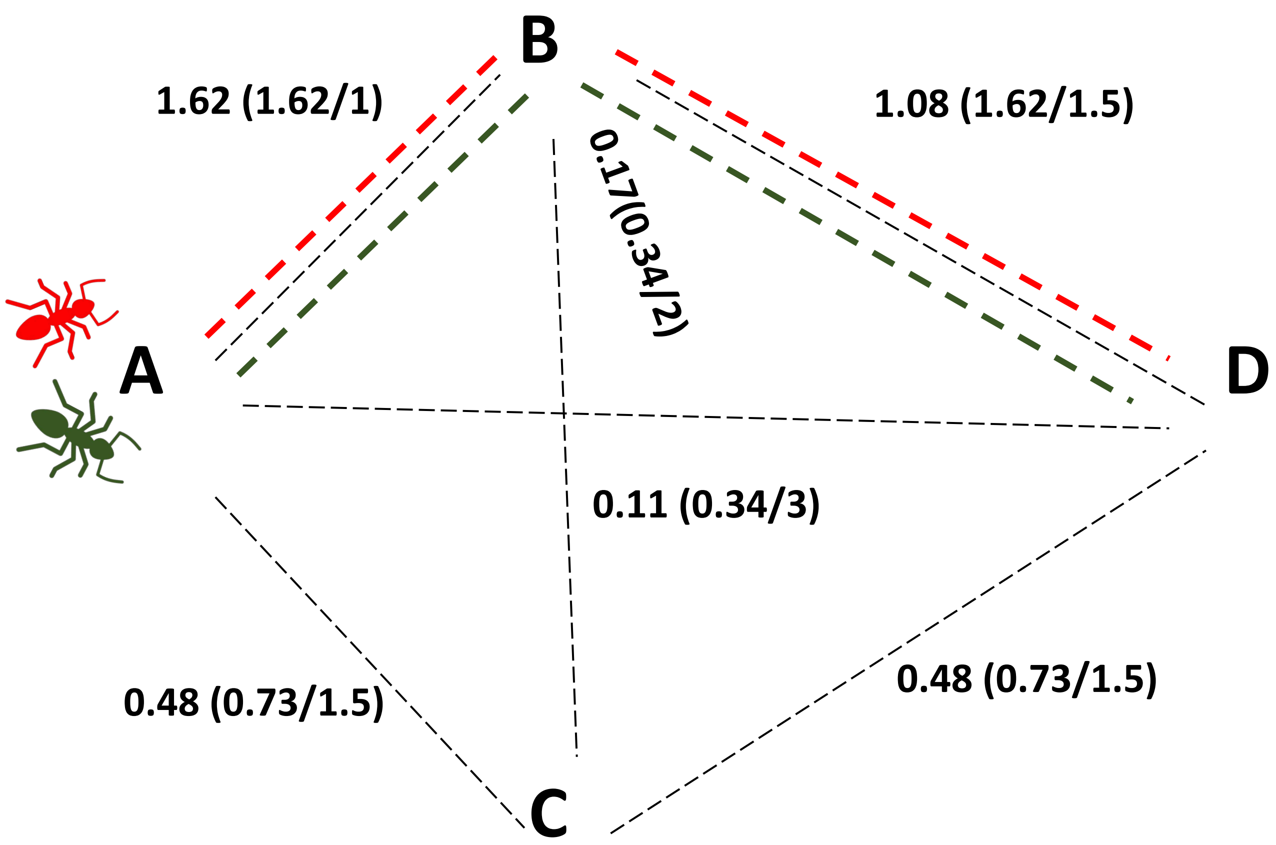
AD: 3; **0.34**; (---)

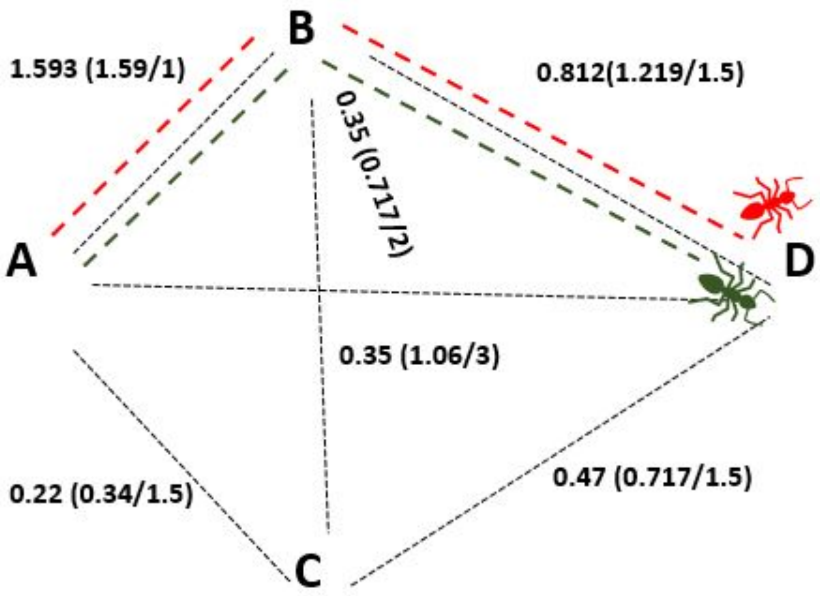
AC: 1.5; **0.73**; (CD)

BD: 1.5; **1.219** (AB, CD)

CD: 1.5; **1.535**; (---)

BC: 2; **0.34**; (AC, AB, BD, CD)





2 Ants
AB,BD
Path length : 2.5
Pheromone deposition : 1/2.5 (0.4)

AB,BD
Path length : 2.5
Pheromone deposition : 1/2.5 (0.4)

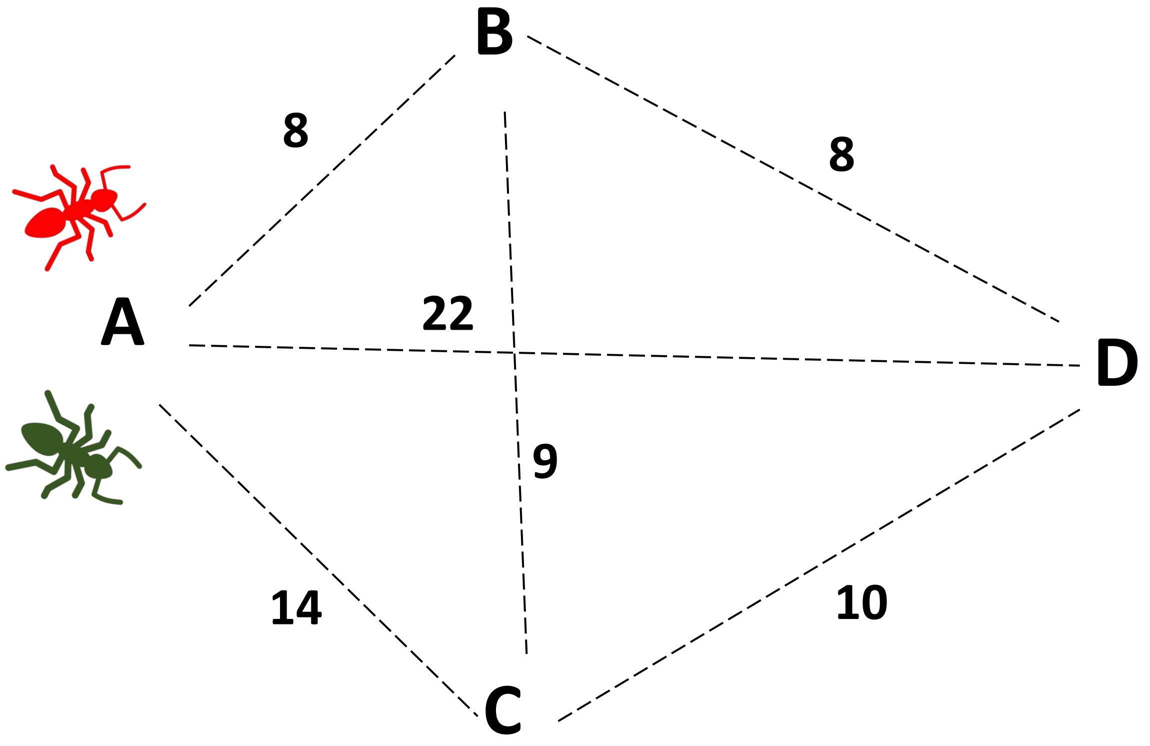
Edges
AB: 1; **1.62**; (BD, BC)
AD: 3; **0.34**; (---)
AC: 1.5; **0.73**; (CD)
BD: 1.5; **1.219** (AB, CD)
CD: 1.5; **1.535**; (---)
BC: 2; **0.34**; (AC, AB, BD, CD)

Edges
AB: 1; 0.7***1.62**+0.4+0.4; (BD, BC)
AD: 3; 0.7***0.34**; (---)
AC: 1.5; 0.7***0.73**; (CD)
BD: 1.5; 0.7***1.219**+0.4+0.4 (AB, CD)
CD: 1.5; 0.7***1.535**; (---)
BC: 2; 0.7***0.34**; (AC, AB, BD, CD)

Edges
AB: 1; **1.93**; (BD, BC)
AD: 3; **0.24**; (---)
AC: 1.5; **0.511**; (CD)
BD: 1.5; **1.65** (AB, CD)
CD: 1.5; **1.07**; (---)
BC: 2; **0.24**; (AC, AB, BD, CD)

INITIAL PHEROMONE LEVELS	FINAL PHEROMONE LEVELS
AB: **1**	AB: **1.93**
AD: **1**	AD: **0.24**
AC: **1**	AC: **0.511**
BD: **1**	BD: **1.65**
CD: **1**	CD: **1.07**
BC: **1**	BC: **0.24**

ACO – SHORTEST PATH PROBLEM

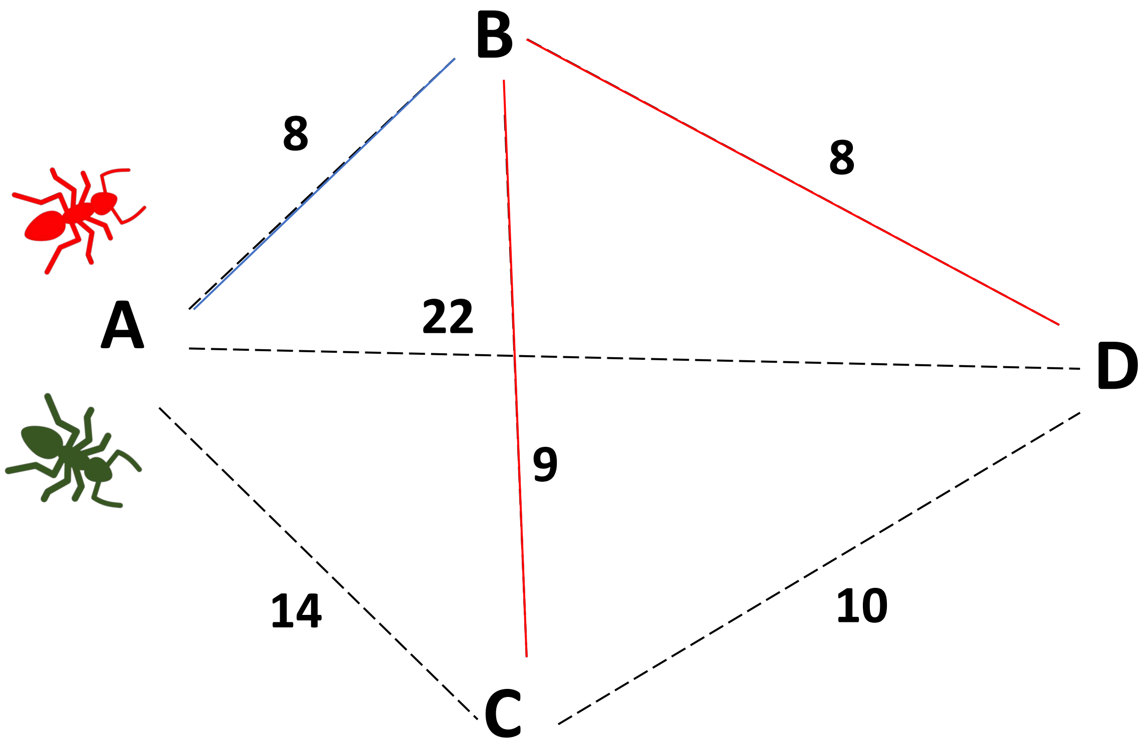


Edges:

AB
BD
AC
CD
AD
BC
CB

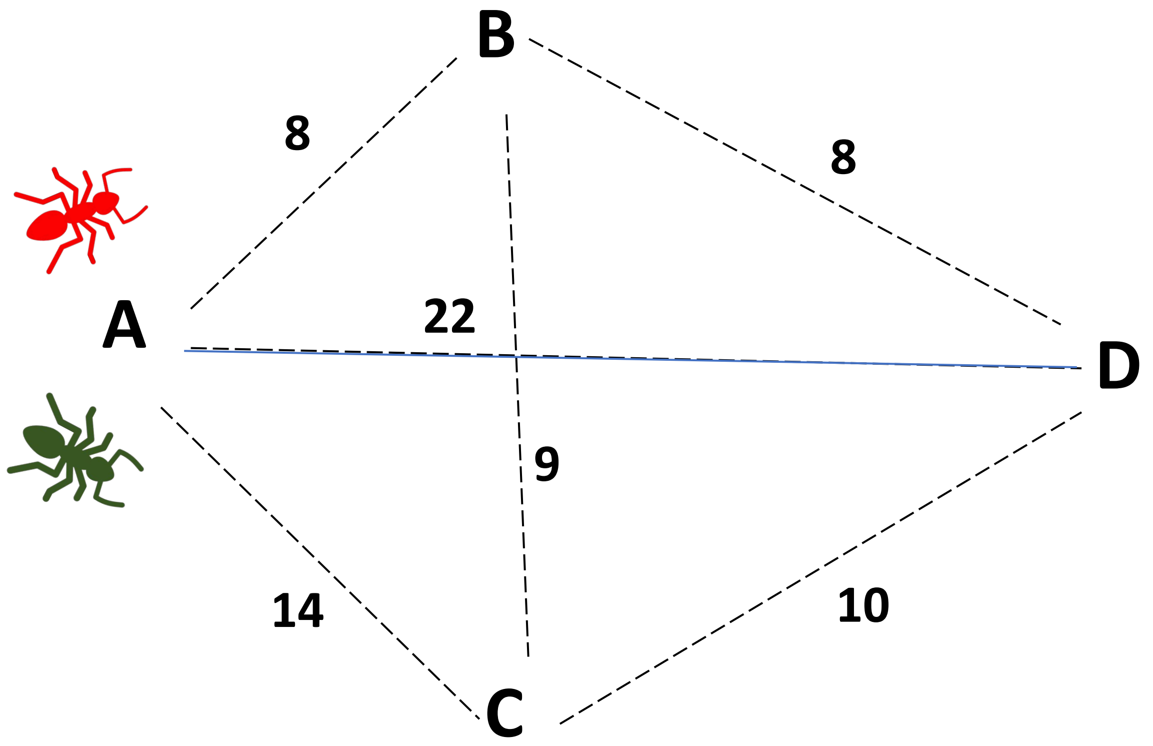
Name
Adjacent
Pheromone conc
Length

ACO – SHORTEST PATH PROBLEM



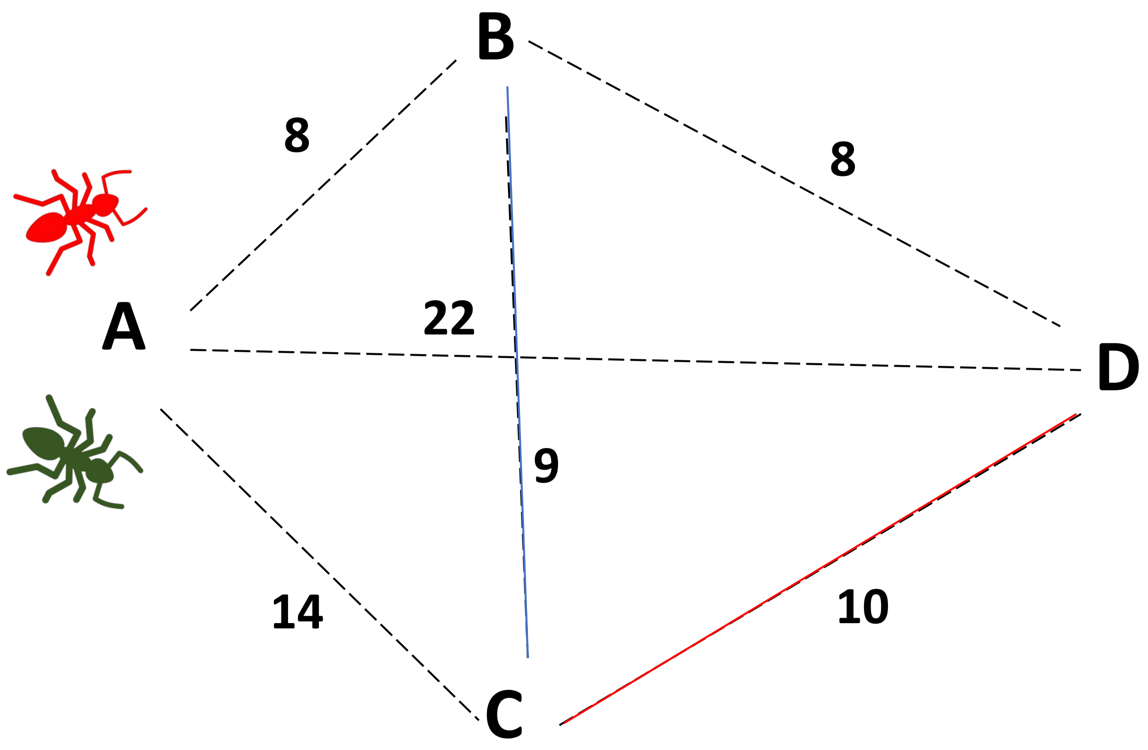
ab = ['AB',['BC','BD'],8,1]

ACO – SHORTEST PATH PROBLEM



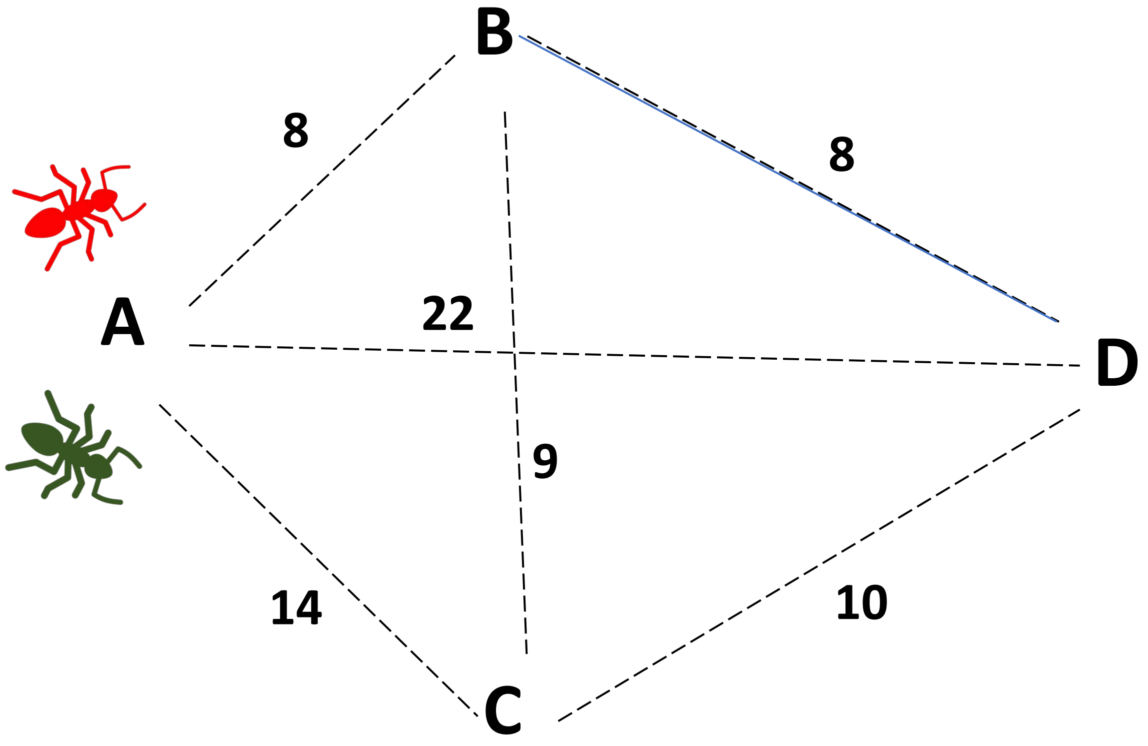
ad = ['AD',[],22,1]

ACO – SHORTEST PATH PROBLEM



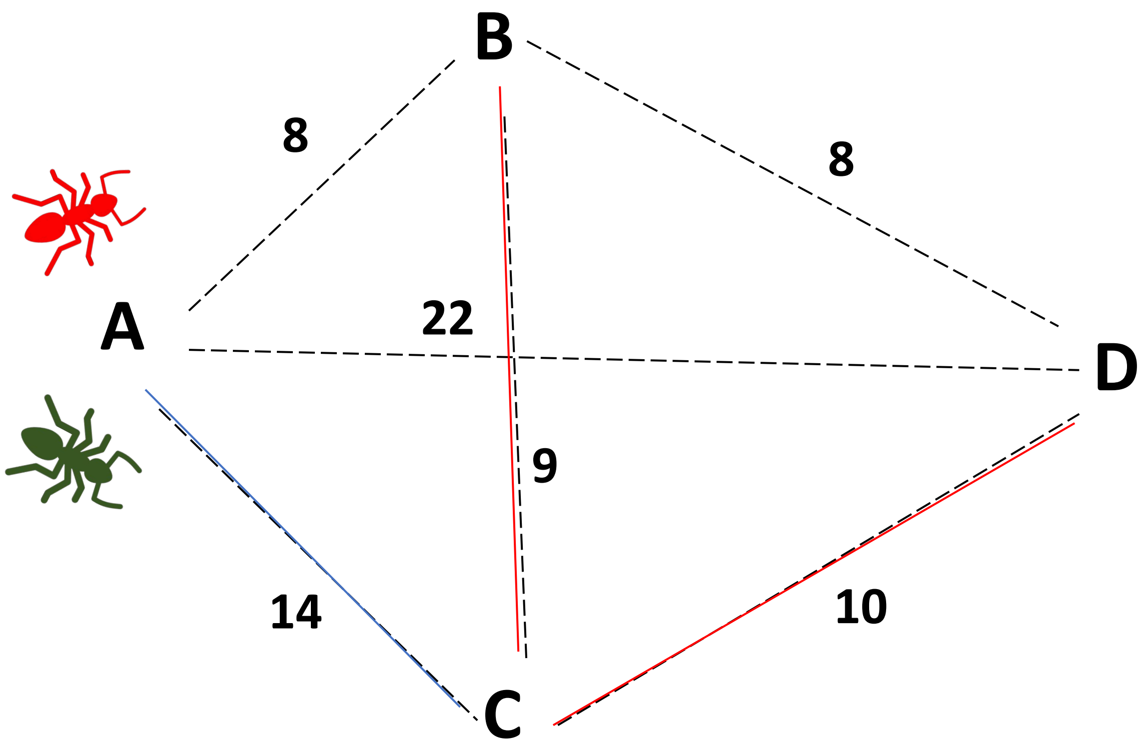
bc = ['BC',['CD’],9,1]

ACO – SHORTEST PATH PROBLEM



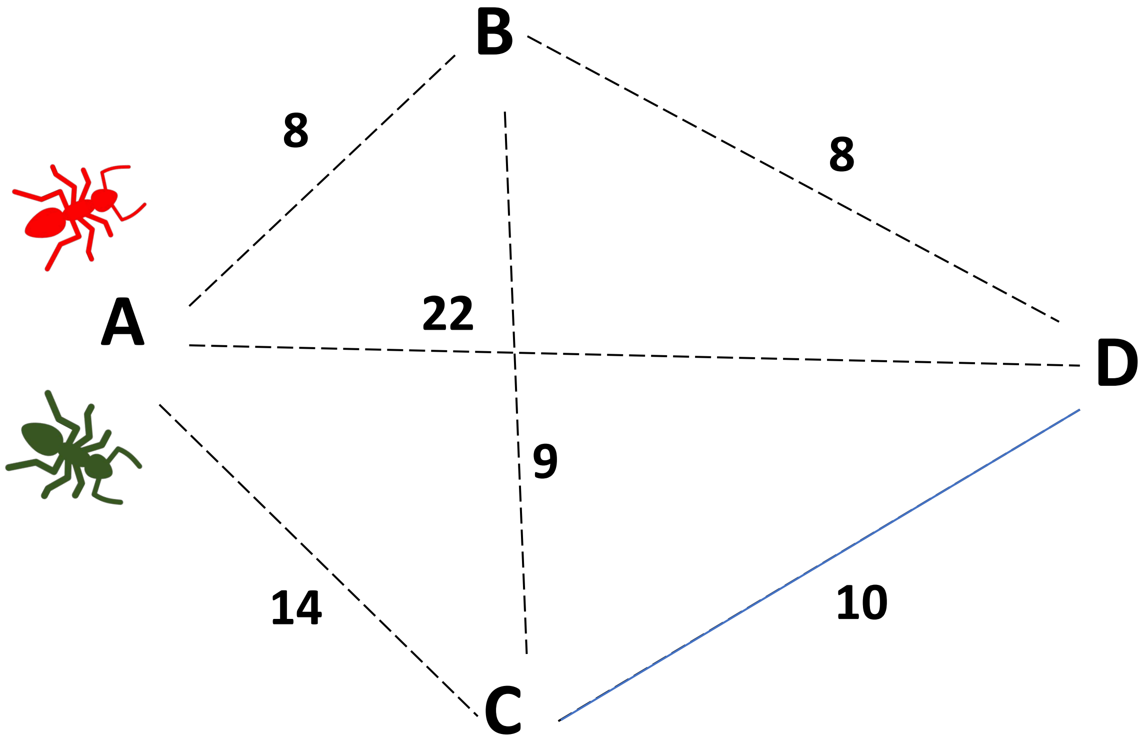
bd = ['BD',[],8,1]

ACO – SHORTEST PATH PROBLEM



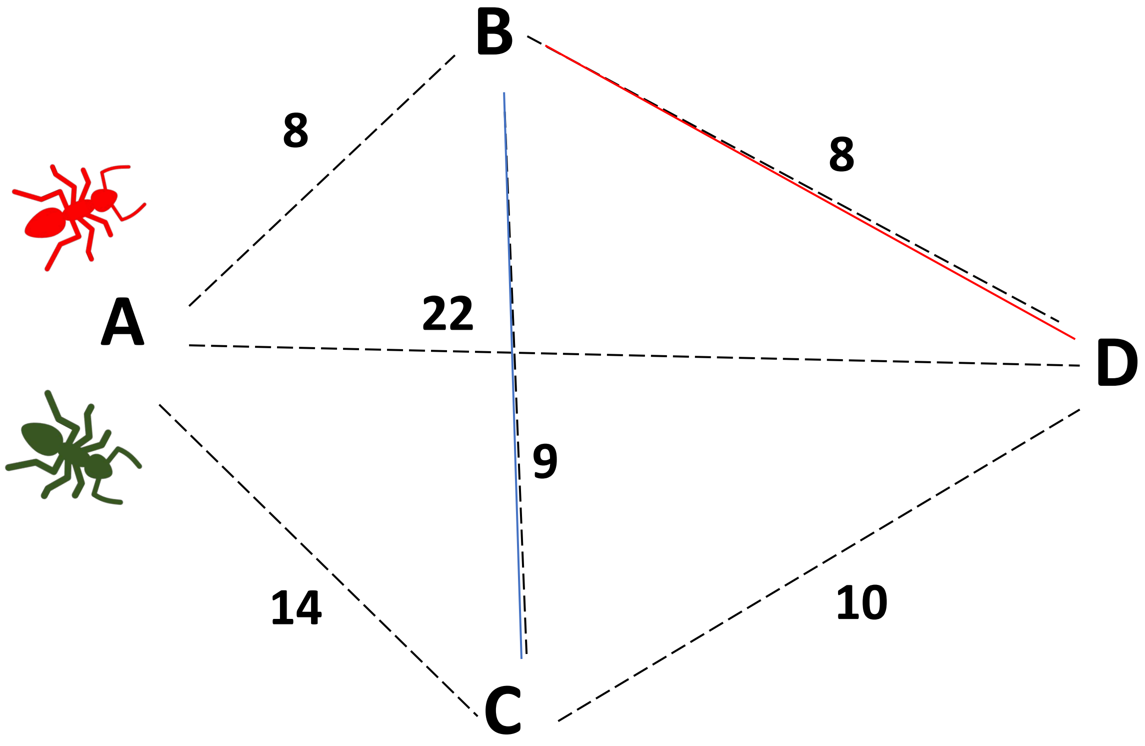
ac = ['AC',[‘CB',’CD'],14,1]

ACO – SHORTEST PATH PROBLEM



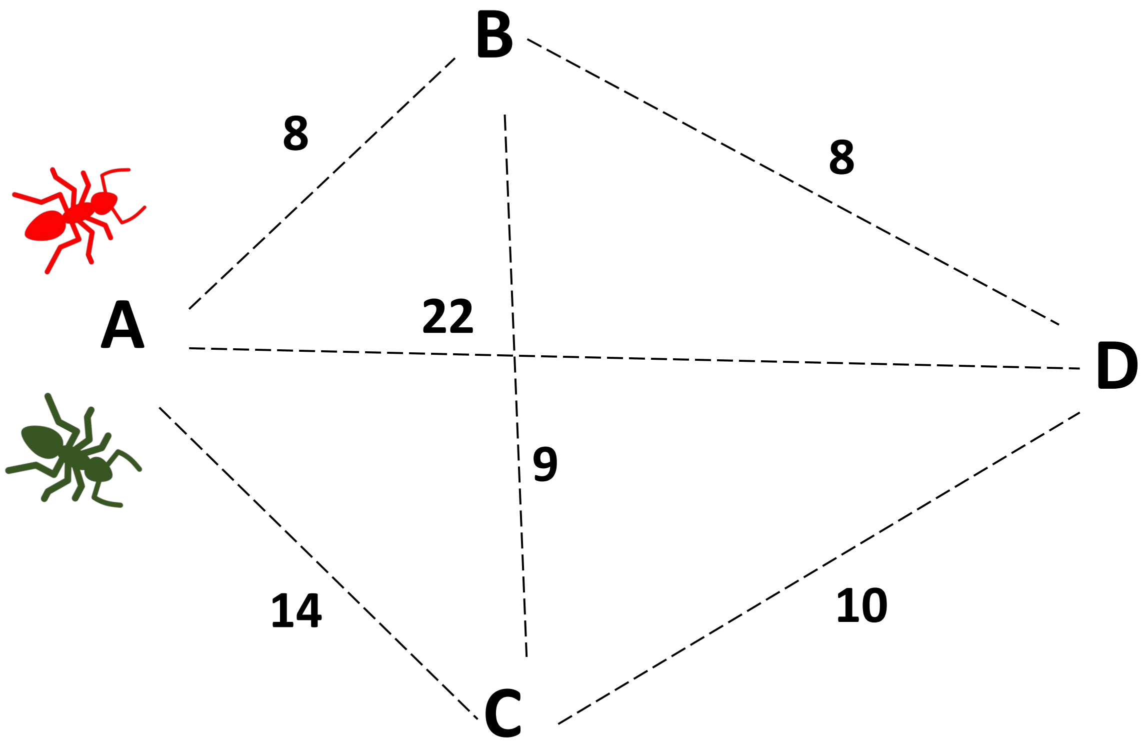
cd = ['CD',[],10,1]

ACO – SHORTEST PATH PROBLEM



cb = ['CB',['BD'],9,1]

ACO – SHORTEST PATH PROBLEM



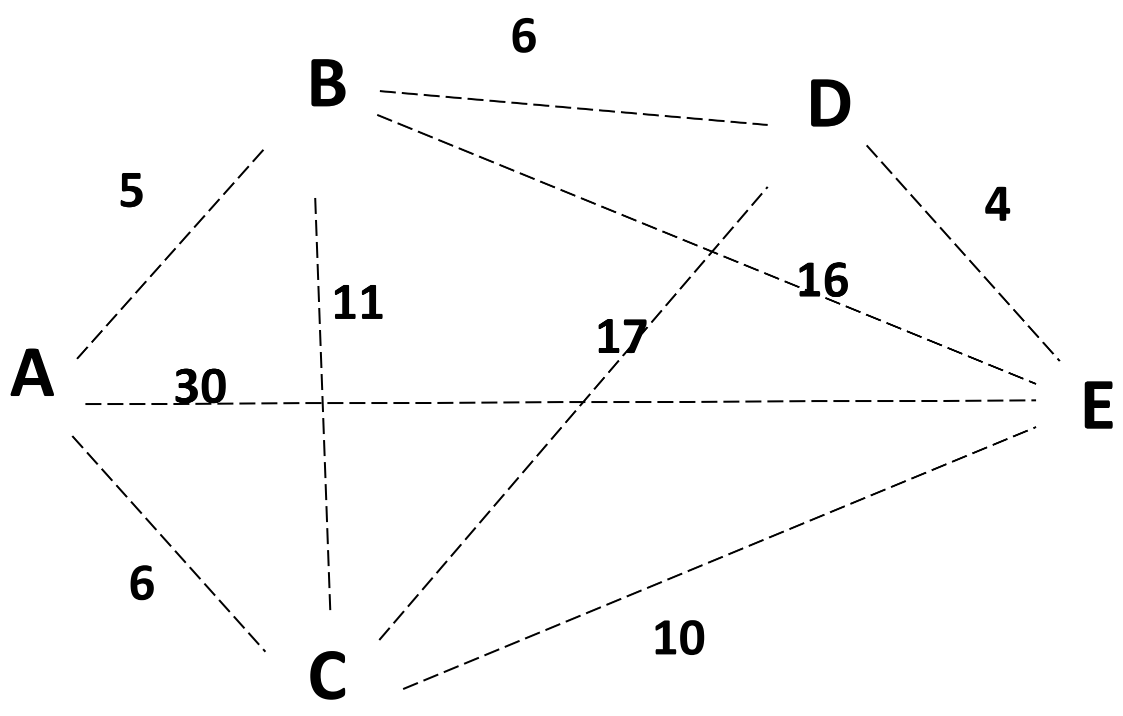
ab = ['AB',['BC','BD'],8,1]
ac = ['AC',['BC','BD'],14,1]
ad = ['AD',[],22,1]
bc = ['BC',['CD’],9,1]
cb = ['CB',['BD'],9,1]
bd = ['BD',[],8,1]
cd = ['CD',[],10,1]

Edge = [name, adjacent edges, length, pheromone level]

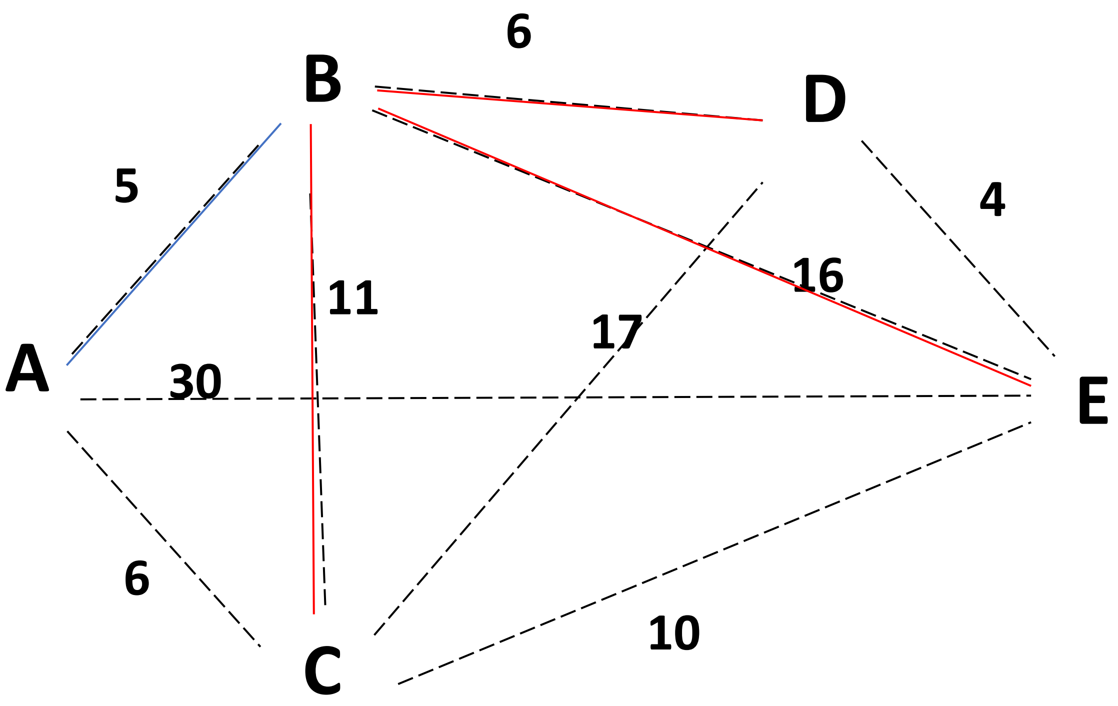
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]

bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```

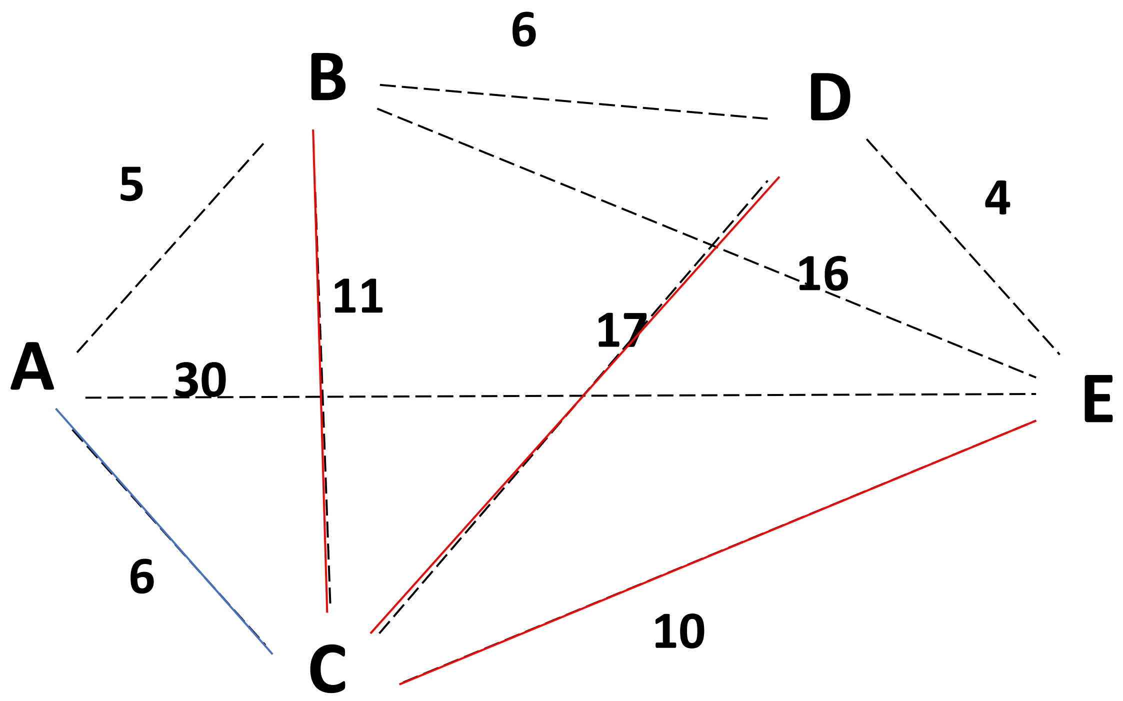
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]



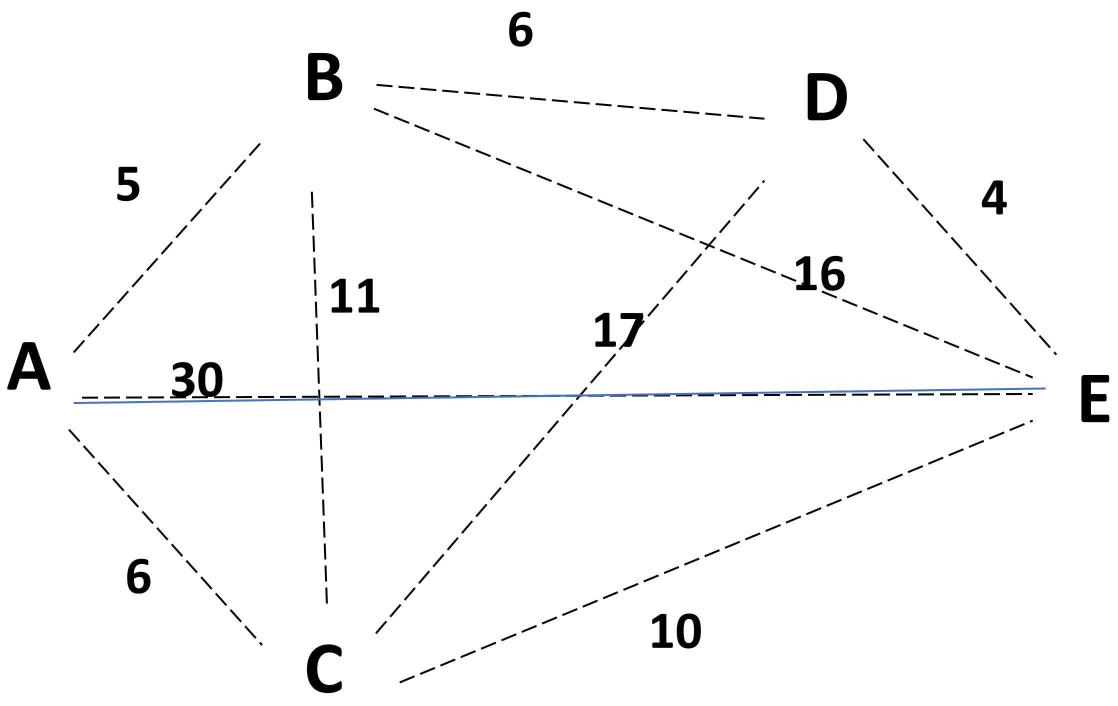
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]



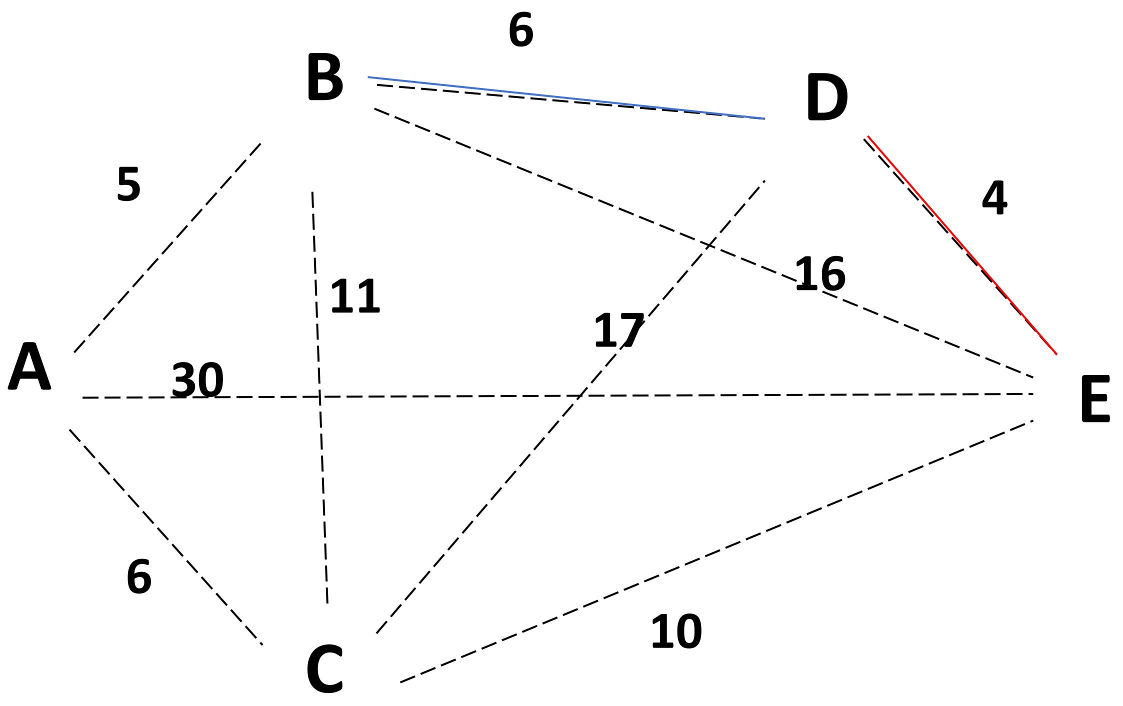
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



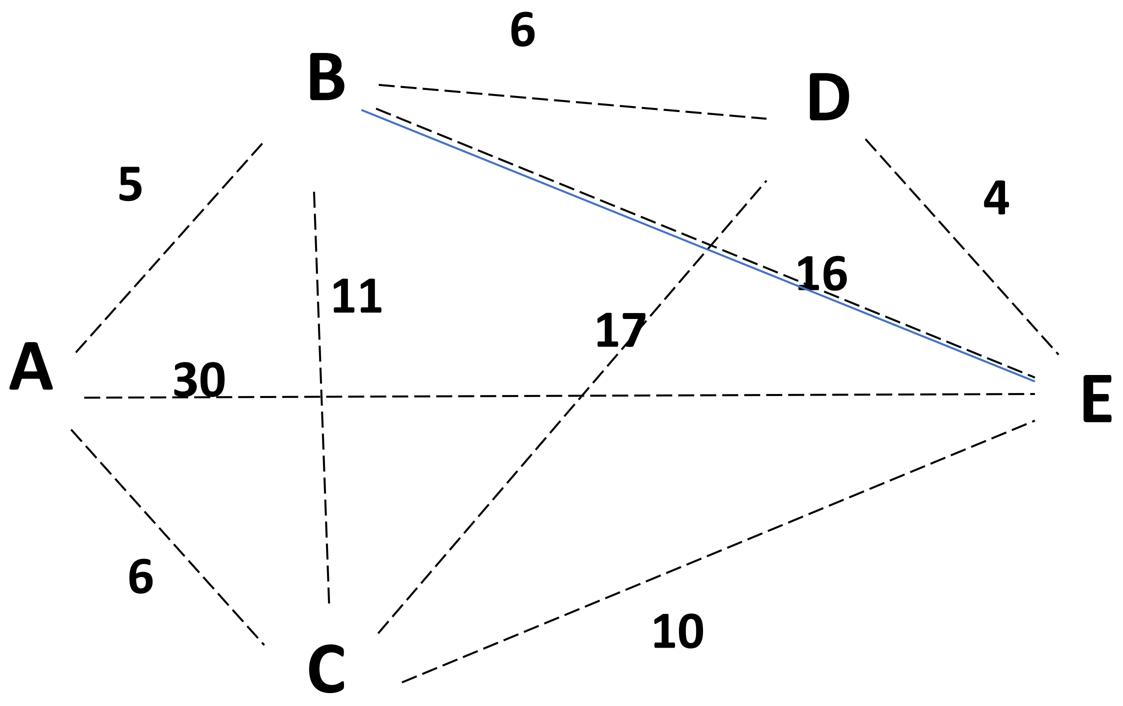
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



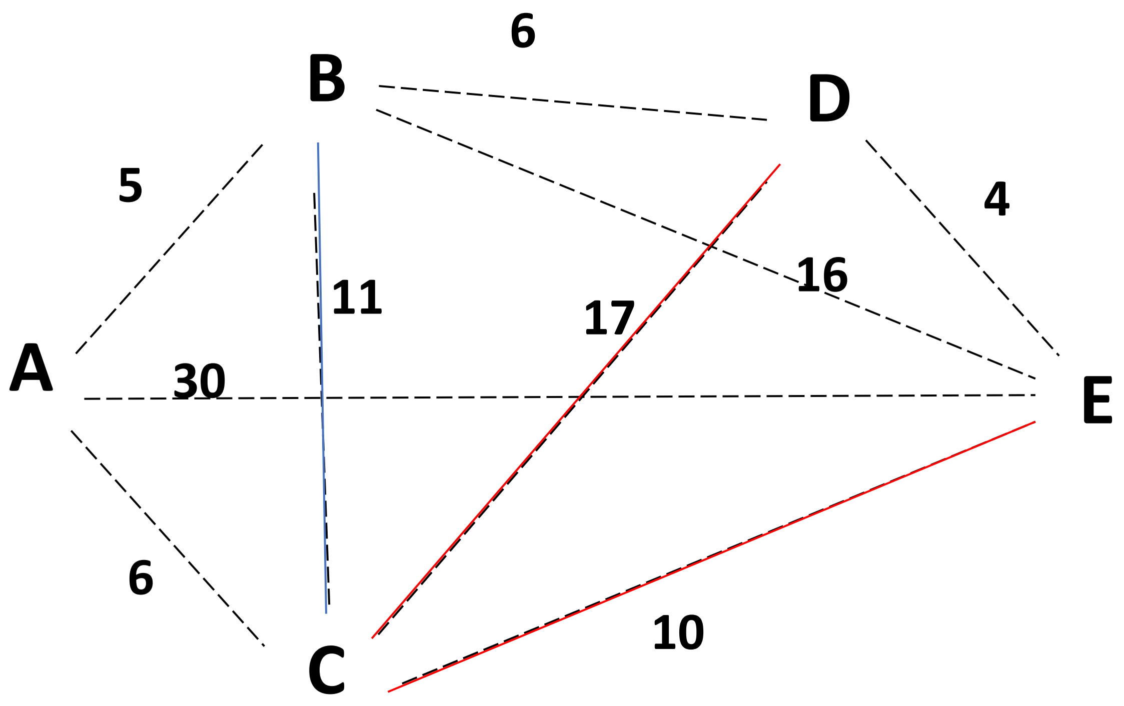
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE’],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]



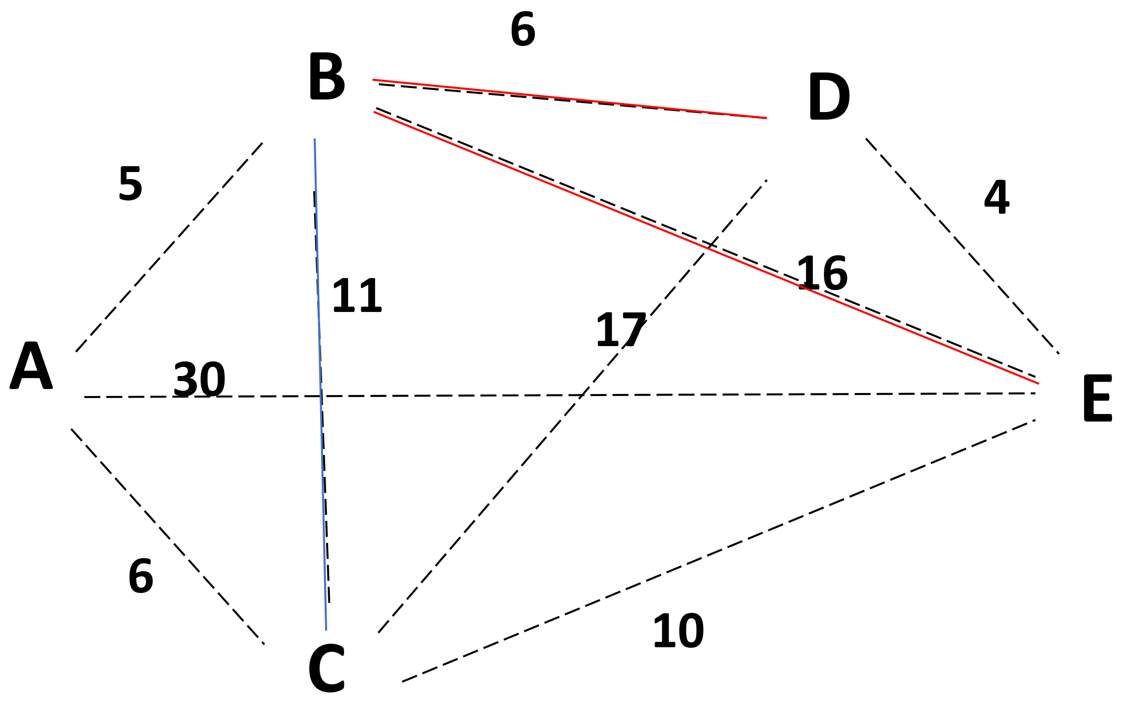
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



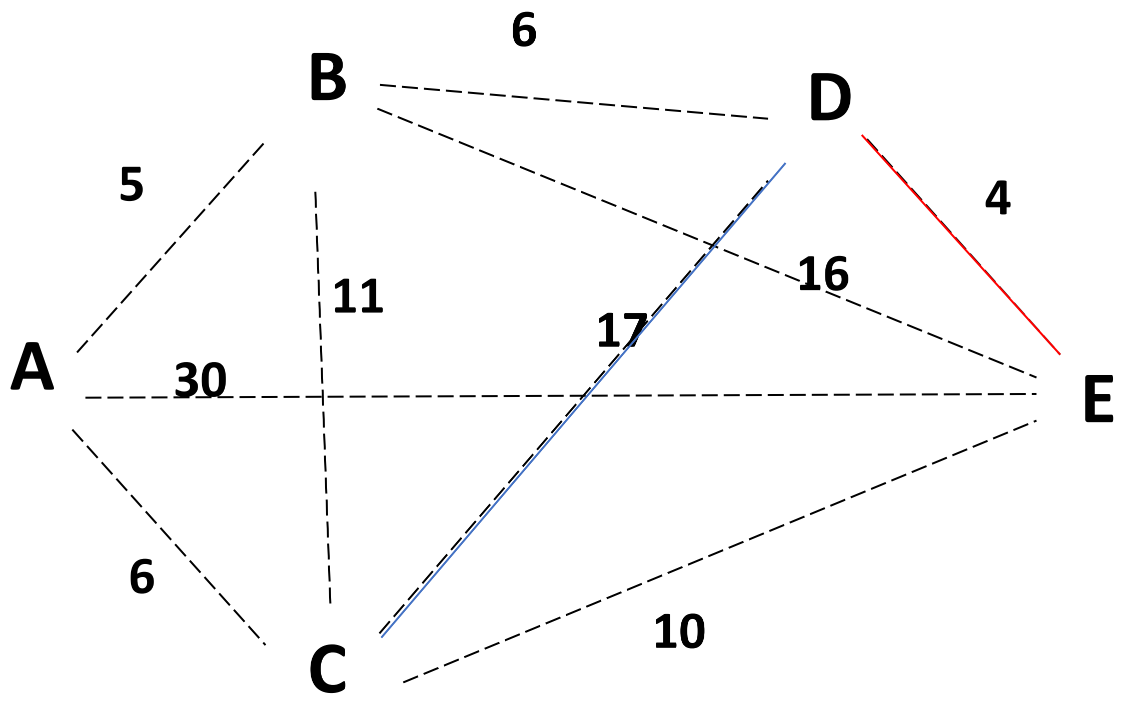
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]



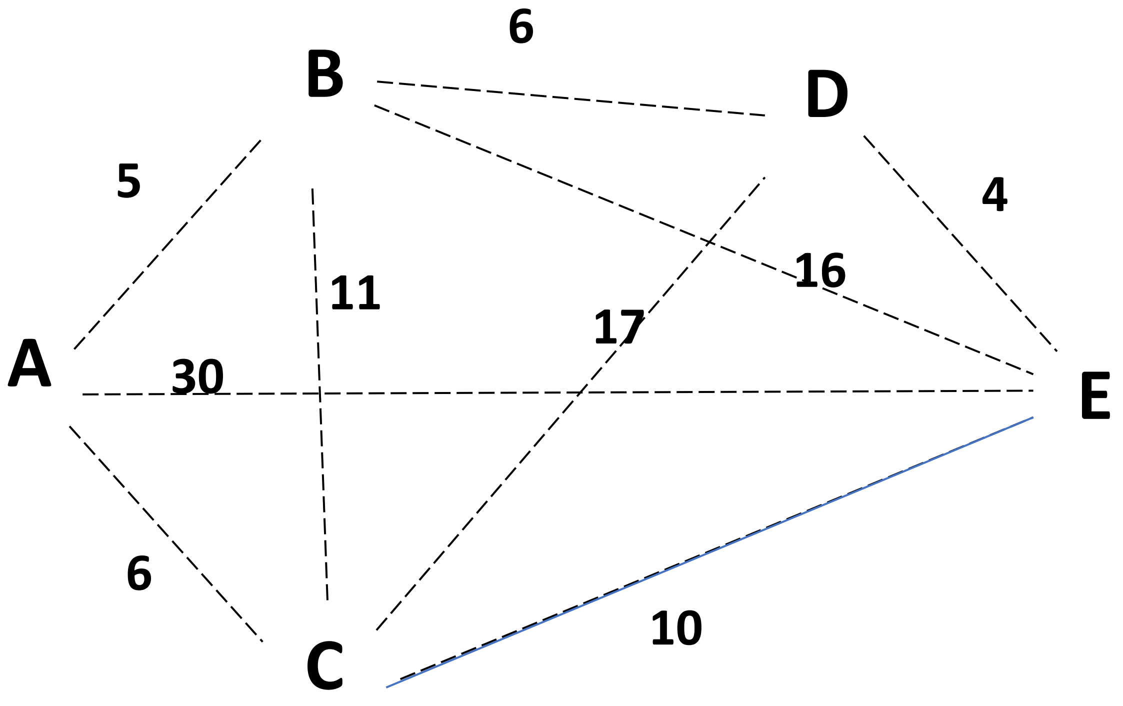
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



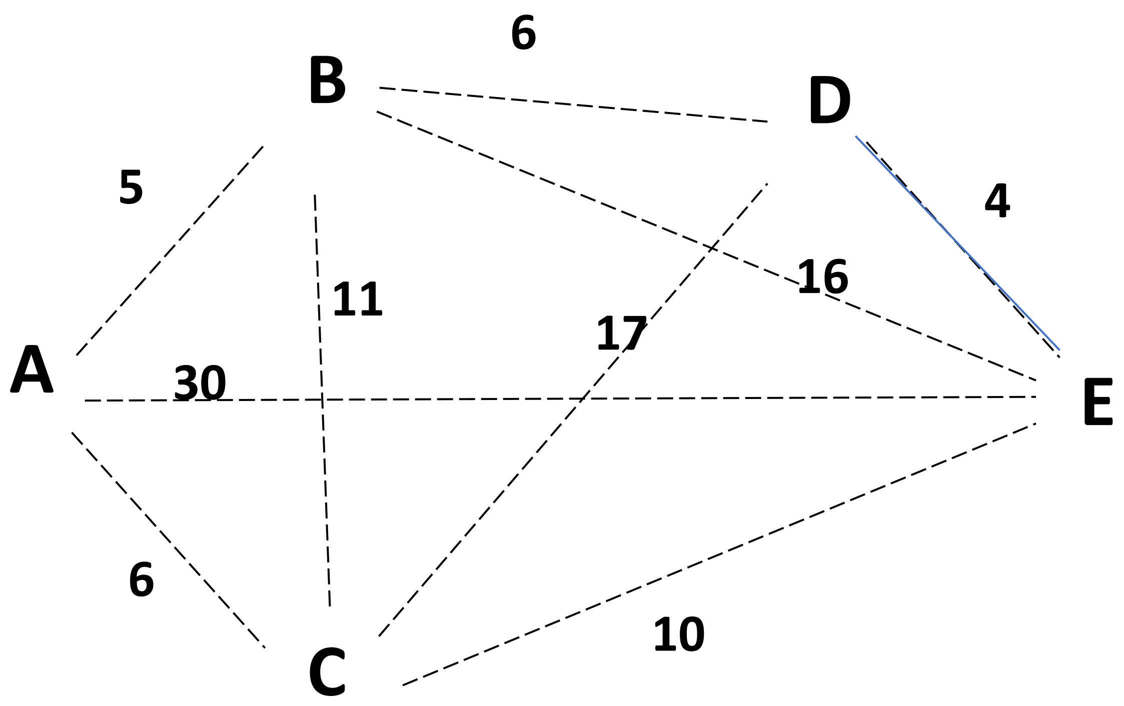
```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```



ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]



```
ab = ['AB',['BD','BE','BC'],5,1]
ac = ['AC',['CB','CD','CE'],6,1]
ae = ['AE',[],30,1]
bd = ['BD',['DE','DC'],6,1]
be = ['BE',[],16,1]
bc = ['BC',['CD','CE'],11,1]
cb = ['CB',['BD','BE'],11,1]
cd = ['CD',['DE'],17,1]
ce = ['CE',[],10,1]
de = ['DE',[],4,1]
dc = ['DC',['CE'],17,1]
```